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Shifting Perspectives

Conducting Visualization Research with Entanglement Epistemology

Derya Akbaba



LINKÖPING UNIVERSITY

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For all the things that bring me joy (and conveniently begin with the letter f): family, friends, food, fig, & fredrik

Acknowledgments

The past is never finished. It cannot be wrapped up like a package, or a scrapbook, or an acknowledgment; we never leave it and it never leaves us behind.

- Karen Barad, *Meeting the Universe Halfway: Preface and Acknowledgments*

Borrowing the words of Barad, this thesis is about *entanglements*. Who am I without the wonderful people in my life with whom I have had the great fortune to become entangled with? Who am I without those who have brought me joy, challenged me to be a better person, cheered me up when I cried, fed me, and taught me things? Who am I without all these people? Entanglement theory, as you will soon read, or perhaps not, answers this question in an elegant way. *Nothing.* We do not exist prior to our relationships nor after. We are influenced by and influence the world around us.

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[1] Will we ever learn who put the cat photos on Floor 3?

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Norrköping, May 2025 Derya Akbaba

Abstract

Foundational theories in visualization offer explanations and models for how people use visualizations to interpret data. A recent turn in visualization research has drawn attention to the ways in which these theories have limited explanatory power. Instead, researchers are drawing on alternative theories of knowledge, or epistemology, to address persistent problems in the field. This dissertation joins the conversation, presenting *entanglement theory for visualization* as an alternative epistemological theory and illustrative case studies that demonstrate how alternative theory in visualization shifts our attention and thus alters visualization research outcomes.

To support this claim, this dissertation presents two contributions that illustrate the productive capacity of feminist theory to contribute to visualization research. The first contribution is a novel theory for visualization transposed from feminist entanglement theory. Entanglement theory for visualization shifts the definitions of data, visualization, and insight toward relational and situated objects that are inseparable from feminist objects of concern: ethics, power, and privilege. Along with new definitions of data, visualization, and insights, we present three ways we mobilize feminist epistemology across visualization as the second contribution. The second contribution consists of three case studies demonstrating how feminist visualization theory drew our attention to previously ignored aspects of visualization design guidelines, the role of knowledge, and ethics in collaborative frameworks. Each case study illustrates the productive nature of approaching visualization research with an alternative epistemology.

Populärvetenskaplig Sammanfattning

Hela världen är fylld av data. Data kan representera många olika saker om våra liv och skapas på flera olika sätt. När vi lyssnar på en låt på repeat dyker den upp igen på Spotify Wrapped bland våra favoriter. I våra bankappar är listan över var vi har spenderat pengar data om våra utgiftsvanor. Väderprognoser skapas utifrån observationer och andra data om vår atmosfär. För att förstå data kan vi visualisera den med hjälp av grafer och diagram. Visualiseringar presenterar data i bilder som gör det lättare för människor att se mönster i datan.

Även om världen är fylld av data är det vi människor som skapar den. Regeringar bestämmer hur de ska samla in folkräkningsdata om befolkningen. Fitnessklockor registrerar vår puls som hjärtfrekvensdata. Till och med att skriva i sin dagbok räknas som data. På samma sätt bestämmer vi människor också hur data ska visualiseras. De som designar visualiseringar väljer både vilka data de vill använda och hur de ska visa dem med färger, linjer och etiketter. På så sätt är både data och visualiseringar produkter av samhället.

Syftet i denna avhandling är att introducera ett alternativt sätt att utforska visualiseringar. För att uppnå detta syfte introducerar avhandlingen ett perspektiv om relationer mellan data, visualiseringar, människor och etik som sedan används i tre fallstudier som visar hur visualiseringsforskning kan bedrivas på ett alternativt sätt.

Den första fallstudien visar hur historien påverkar dagens forskning kring riktlinjer för minimalistisk design. Den andra fallstudien visar hur det nya perspektivet kan ge upphov till nya designidéer för kollaborativa visualiseringsverktyg. I det sista exemplet diskuterar vi resultaten från en intervjustudie med visualiseringsforskare och de forskare som de samarbetade med. Utifrån resultaten föreslår vi nya sätt att samarbeta med fokus på omsorgsetik. Denna avhandling visar att det är möjligt att bedriva visualiseringsforskning som utgår från perspektiv på relationer, makt och samhälle. Att reflektera över hur vi skapar och visualiserar data öppnar upp för ny forskning och skapar ny kunskap om hur visualiseringar används i samhället.

Publications

The following list of publications have been included in this thesis (presented in chapter order):

Paper I: D. Akbaba, L. Klein, and M. Meyer. Entanglements for visualization: Changing research outcomes through feminist theory. *IEEE Transactions on Visualization and Computer Graphics*, 31(1):1279–1289, 2025. doi:10.1109/TVCG.2024.3456171

Paper II: D. Akbaba, J. Wilburn, M. T. Nance, and M. Meyer. Manifesto for putting 'chartjunk' in the trash 2021! *alt.VIS, a workshop co-located with IEEE VIS, 2021*, 2021. doi: 10.48550/arXiv.2109.10132

Paper III: H. Lin, D. Akbaba, M. Meyer, and A. Lex. Data hunches: Incorporating personal knowledge into visualizations. *IEEE Transactions on Visualization and Computer Graphics*, 29(1):504–514, 2023. doi:10.1109/TVCG.2022.3209451

Paper IV: D. Akbaba, D. Lange, M. Correll, A. Lex, and M. Meyer. Troubling collaboration: Matters of care for visualization design study. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*, CHI '23. Association for Computing Machinery, 2023. doi:10.1145/3544548.3581168

Paper V: D. Akbaba and M. Meyer. "Two heads are better than one": Pair-Interviews for visualization. In *2023 IEEE Visualization and Visual Analytics (VIS)*, pages 206–210, 2023. doi:10.1109/VIS54172.2023.00050

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CHAPTER 1

Introduction

This dissertation *transposes* — interprets and operationalizes — feminist epistemological theory for visualization research to offer a different way to study how visualizations work in the world. This work comes at a time when there is a rise of new theories to address persistent problems, but only a handful of literature questioning the field's foundations by interrogating visualization's epistemology. [13, 23, 34, 35, 36, 42, 96, 107, 116]. The rise of new theory in visualization is not systematic nor coordinated, but rather an organic, emergent discussion. Some have mobilized Bayesian statistics to explain the effects of beliefs and prior knowledge on visualization readers [82, 83]; others have turned to behavioral economics as explanatory theories of decision making [78]; and some interpret learning science to highlight the pedagogical and communicative goals of visualization design [1, 94]. This dissertation contributes to the conversation by articulating a novel theory for visualization that engages with feminist concerns of power and privilege — *entanglement theory for visualization* — and illustrative case studies of the theory in practice.

Entanglement theory for visualization reframes familiar visualization concepts — data, visualization, and insight — as relational and situated objects, shifting research focus to a different set of concerns informed by feminist entanglement theory [7]. Additionally, toward the dissertation's aim, we present illustrative case studies that demonstrate the productive capacity of feminist theory for visualization research across design guidelines, the role of knowledge in visualization design, and ethics in collaborative frameworks. The first case study reviews a common minimalist design guideline and the rhetorical harm its use perpetuates. Our second case study demonstrates how theory can alter design constraints, resulting in a new visualization design. And our final case study presents an interview study that challenges normative collaboration frameworks and suggests ethical alternatives. With this dissertation, we demonstrate a different way to conduct visualization research and the possibilities that follow.

1.1 Motivation

There are already many theories and models of how visualizations work in the world. These *foundational theories* — originating from engineering, perceptual, and cognitive sciences — have

driven much of the early work in visualization. Theories from perceptual sciences build on empirical evidence that explain how to design a visual display so that the visualization amplifies an individual's cognition [57]. Such theories provide guidance on how to design visualization interfaces that support accurate and precise decoding by the user. Important theories in this area contribute to the ranking of visual channels and marks [32, 113], enabling the field to make claims like *position is easier to decode more precisely than area*. Theories of cognition and perception also guide color choice, catalog optical illusions, and suggest appropriate use cases for animation [57]. Yet, despite numerous foundational theories, it remains a persistent problem that design guidelines still fail to fully explain how visualizations work.

Several recent studies reveal the shortcomings of foundational theory in visualization, including the lack of theorization about people beyond their perceptual abilities, data as created objects, and visualizations as non-neutral representations. In a study in rural America, researchers found that people engaged more with visualizations that depicted topics with which they had a personal connection to [123]. In scientific collaborations, researchers have reported on the challenges of creating useful visualizations for their collaborators, either because the data — and consequently the visualization — could never represent reality [96, 104, 120] or other socio-technical factors that meant the visualization could no longer support the researcher [4]. And despite the substantial research on how to visualize uncertainty, Hullman reports that many designers choose not to communicate uncertainty [73]. There are numerous examples of research that challenge and trouble the foundational theories of visualization, signaling that an *epistemic wall* [5] — the boundary of explainability of any one epistemology — has been reached.

In this dissertation, we suggest that it is time for the community to explicitly discuss and create epistemological theories for the field. Epistemology is the study of how we know what we know. In other words, it is how we create and validate new knowledge in research. The field of visualization orbits questions of knowledge: how to leverage existing knowledge [29]; how to gain new knowledge through visualizations [132]; and how to communicate knowledge [102]. And yet, for a field so interested in knowledge, there is hardly anything written about epistemology. Different epistemological stances *shift* researchers' attention and perspectives, fundamentally altering how science is conducted and knowledge is validated. And so, discussing epistemology is a more precise way to both discuss the challenges we face theoretically, but also to scale epistemic walls and explore new paths for research.

This dissertation presents an epistemological theory for visualization that draws explicitly on feminist epistemology. Feminist theory has found its place in visualization research as a response to the growing awareness that visualizations are more persuasive than neutral and can perpetuate bias and other harmful stereotypes. Feminist and other critical visualizations explicate the ethical considerations of designing visualizations [42, 44]. This line of research has made significant strides in the community by introducing terms from feminist scholarship and drawing attention to underdeveloped aspects of foundational theories from the perspectives of power and privilege, including: researcher ethics, theorizations of data, and the rhetorical power of visualization. While this prior work interprets and presents solutions for visualization researchers, it does not generate new epistemological theory for visualization.

This dissertation presents entanglement theory for visualization, contributing a new epistemological theory for visualization. Additionally, we present a series of case studies that illustrate the productive capacity of conducting visualization with a different epistemology. This dissertation serves as another voice in the ongoing discussion by visualization researchers seeking to overcome hurdles in their research by applying new methods and theories from other disciplines. Together with others, this dissertation contributes to and advances epistemic plurality by introducing feminist theory for a visualization research context, contributing both theoretically and practically through a series of projects that showcase the productivity of this epistemological stance.

1.2 Contributions

The primary contribution of this dissertation is advancing epistemology within a visualization research context. The contribution lies on two pillars: theory and practice. As the first pillar, we present entanglement theory for visualization, a novel theory that outlines feminist entanglement theory for visualization. As a theory, it comprises 1) the articulation of how data, visualization, and insight are studied in the context of the social, historical, and technical conditions that produce them; and 2) the set of ethical implications that follow a feminist stance on knowledge production. The second pillar, practice, demonstrates the application of this theory across three case studies covering design guidelines, visualization design, and collaboration. These separate case studies illustrate the productive capacity of engaging with feminist theory for visualization research.

This dissertation includes several secondary contributions that follow the case studies. We present a set of provocations that suggest alternative ways to collaborate; new visualization methods for conducting interviews, analyzing interview transcripts, and performing maintenance art; and a new conceptualization of qualitative uncertainty and design guidelines for incorporating this concept into a collaborative visualization tool. The primary and secondary contributions are drawn from the following previously published work:

> Derya Akbaba, Lauren Klein, and Miriah Meyer. *Entanglements for Visualization: Changing Research Outcomes through Feminist Theory.* IEEE Transactions on Visualization and Computer Graphics (Proceedings of VIS 2024), 2025.
>
> Derya Akbaba, Jack Wilburn, Main T. Nance, and Miriah Meyer. *Manifesto for Putting 'Chartjunk' in the Trash 2021!* alt.VIS, a workshop co-located with IEEE VIS, 2021.
>
> Haihan Lin, Derya Akbaba, Miriah Meyer, and Alexander Lex. *Data Hunches: Incorporating Personal Knowledge into Visualizations.* IEEE Transactions on Visualization and Computer Graphics (Proceedings of VIS 2022), 2023.
>
> Derya Akbaba, Devin Lange, Michael Correll, Alexander Lex, and Miriah Meyer. *Troubling Collaboration: Matters of Care for Visualization Design Study.* ACM SIGCHI Conference on Human Factors in Computing Systems, 2023.
>
> Derya Akbaba and Miriah Meyer. *'Two Heads are Better than One': Pair-Interviews for Visualization.* Proceedings of IEEE VIS (Short Papers), 2023.

1.3 Organization

This dissertation summarizes related work about theory in visualization and then presents the contributions, beginning with entanglement theory for visualization, followed by three case studies. To situate the contributions, in Chapter 2 we define relevant terminology, provide a condensed history of foundational theories in visualization, and review new research that mobilizes alternative theories within visualization research. This chapter lays the groundwork for why visualization researchers should consider and discuss epistemology in their work. In Chapter 3 we introduce feminist entanglement theory, its corresponding epistemological and ethical considerations, and the theory that transposes these considerations into a visualization context. The chapter ends with a set of core concepts that are mobilized and illustrated in the subsequent case study chapters.

The following three chapters of this dissertation are a series of case studies that showcase *how* engaging with theory ultimately changes visualization research outcomes. In Chapter 4, we present our work that examines minimalist design guidelines through a feminist entanglements perspective, demonstrating the rhetorical harms that fall from using the concept *chartjunk* as a serious research topic. In Chapter 5, we describe novel design guidelines that embody feminist theories of data and knowledge. In Chapter 6, we present our interview study with visualization researchers and their collaborators to demonstrate how feminist ethics troubles extant frameworks for collaborations. In Chapter 7, we summarize the dissertation and conclude with a discussion of future work.

1.4 Chapter Formatting

For legibility and continuity, the core chapters (Chapters 3–6) follow similar formatting. Each chapter begins with a short summary of the chapter, along with the names of the published works that the chapter builds on. Following this, there is a brief *backstory* [1] about the research. The backstory describes my specific contributions within the project's research team and provides context for how the ideas came to be. The goal here is to illustrate the influences that shape research, building on the core epistemological claims of the thesis, that it's all entangled.

Afterward, I present an overview of the published research [2] that serves as the body of the chapter. Each chapter ends with a section called *reflections for dissertation* which serves as a summary of the chapter and a reflection about the work as it relates to the core contributions and claims of the dissertation.

[1] The term backstory is borrowed from Yang et al.'s paper titled *The Backstory to "Swaying the Public": A Design Chronicle of Election Forecast Visualizations* [163]. It is an excellent contribution to the visualization community as it demonstrates the complicated and often unclear paths that research can take.

[2] The curious reader is welcome to read the published articles, which are reprinted with permission in this dissertation.

CHAPTER 2

Related Work

Epistemology is the branch of philosophy concerned with knowledge and how knowledge is produced. Understanding knowledge practices involves examining the standards and values used to shape and evaluate knowledge claims. In other words, epistemology is how we know what we know about the world. Traditionally, visualization research has been conducted from a single epistemic perspective. However, speaking precisely about epistemology in visualization is difficult because it is implicit rather than explicit across research outputs.

This dissertation takes the position that early visualization research has followed a similar intellectual beginning to that of the human-computer interaction (HCI) field. As such, we borrow the language from HCI researchers [16, 59, 64, 65] and recent visualization research [107] to ascribe a loose epistemology to normative visualization research. These theoretical foundations are rooted in a type of *positivism* that emulates the intellectual traditions of engineering and cognitive science. Positivism posits that knowledge claims are objective and neutral, produced in a manner that strives to remain unbiased and unaffected by the individual conducting the research. In the following section, we expand upon the theoretical foundations of visualization, calling attention to how positivism has shaped visualization research thus far.

2.1 Theoretical Foundations of Visualization

Epistemology can be interpreted as what one pays attention to during the process of knowledge generation. In visualization, the field's attention has been focused on *insight* as articulated by a well-known saying within the community: *"The purpose of visualization is insight, not pictures"* [29, p. 6]. The authors responsible for this saying formalized the idea into a model known as the *reference model*. The reference model (see Fig. 2.1) is *"the mapping of data to visual form that supports interaction in a workspace for visual sense making"* [29, p.17]. This definition establishes that a visualization begins with *raw data* that is turned into visual structures that enable an individual to iteratively explore the data visually. It is also important to note that this early model bounds visualization use to an analytical context. Keim et al. extend the model to further specify component parts and behaviors during visual analytics [132]. And

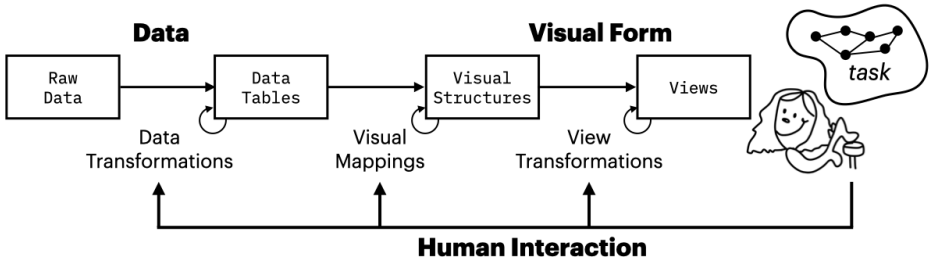


Figure 2.1 — Author's rendition of the reference model [29]. The model represents the transformation of data to visual form through a series of human-initiated interactions. These interactions include data transformations, visual mappings, and view transformations.

Munzner uses this model as the basis of a process model for visualization design and evaluation more generally [112]. Undergirding these models are the cognitive and perceptual theories that put forward rankings of visual channels along metrics of perceptual effectiveness and efficiency. Additionally, across these models are overlapping and important component parts central to the theoretical foundations of visualization — visualization, data, and insights — which are discussed throughout the rest of this section.

An important aspect of the foundational theory in visualization is the idea that visualizations are tools to amplify cognition. The design of these tools has followed a minimalist design aesthetic that can be traced through the most commonly presented historical account of visualization. The lineage of visualization often begins with William Playfair as the earliest person to represent data visually. [1] Following Playfair, Jacques Bertin formalized a theory of graphics in *The Semiology of Graphics* [14]. Another influential person was John Tukey, who introduced visualization into the statistical data analysis pipeline. In his work *Exploratory Data Analysis*, he brought forward the idea that visually representing statistical information could support data analysis [146]. These ideas are later brought to the public sphere in a series of self-published and highly popular books by Edward Tufte [144, 145]. In these books, Tufte declares that scientific visualizations should follow minimalist design standards to uphold *graphical integrity* [2]. From here, Cleveland & McGill contribute *Dynamic Graphics for Statistics* that illustrates how visualizations can support the analysis of multidimensional data with many variables [31]. These ideas from statistical graphics champion minimalism and clarity; further, they are driven by the shape and attributes of the data [125, 155], which reinforces a core foundational theory that visualizations are windows

[1] In this section, I am presenting both an abridged history and one that follows a normative telling of visualization as presented by Card and colleagues [29]. This is partly because this is what most students still learn in their introductory courses, but also to emphasize that, as a community, we tend to draw on the same handful of examples. For a more in-depth review of visualization history from a Western and statistics-influenced perspective, I point you to *A History of Data Visualization and Graphic Communication* [60] and *History of Information Graphics* [129]. As an alternative historical account, I recommend *Data by Design* as a retelling of visualization history from a more critical perspective that draws on early visualization work led by women and people of color [86].

[2] See Chapter 4 for more historical context about Tufte and his impact on the visualization community.

to the data. Both this philosophy of visualization and design are readily seen in present-day visualization guidelines and programming libraries (e.g. d3.js, ggplot, VegaLite). This brief historical account of visualization highlights that statistics and positivist epistemology shaped a very specific conceptualization of visualization: one that is data-driven and requires little design flourish to faithfully represent the data.

Another central idea driving the theoretical foundations of visualization is that visualizations help an analyst see patterns in the data, otherwise known as *insights*. Although many have tried to define insight, there is still no agreed-upon definition [10, 79]. Despite a lack of consensus, a shared objective across researchers has been to quantify and measure insight to evaluate the effectiveness and value of the visualization [114, 124, 139, 164]. An early example is the *insight-based methodology* put forward by Saraiya and colleagues [134]. In this methodology, they assign scores to insights based on their value to the domain. Other researchers have suggested alternative methods for recording and quantifying insights in order to assess and compare visualizations [66, 138], with some going so far as to suggest using EEG or fMRI [30]. Insights are measured because a core theoretical foundation of the field is the idea that it is possible to optimize the efficiency and effectiveness of visualization and thus objectively know which visualization designs are better than others for a given task. This rests upon theories of cognition from psychology and cognitive science that make it possible to test and rank the effectiveness of visual marks.

Fundamental theories in visualization about cognition are grounded in empirical evidence from cognitive science, psychology, and perceptual studies. These theories provide guidance for how and when to use specific visual channels over others [57, 113]. These guidelines are intended to support visualization design that leads to predictable and accurate decoding of data values. To this aim, studies have developed theories for increasing chart legibility, reducing deception, conveying uncertainty, and supporting accessibility across color-vision impairments. For a more thorough review of perceptual and cognitive theories employed across visualization research, we refer you to the work by Franconeri and colleagues [57].

In sum, the theoretical foundations of visualization are a set of theories and models about how visualizations amplify cognition so that people can more easily see patterns in data. While the specific epistemology of these theories is not explicitly discussed, there is a general consensus and acceptance of these foundational theories. However, recent research has challenged the underlying assumptions of this foundational work. In this dissertation, we argue that in some aspects of visualization research, we have hit an *epistemic wall*, or the boundaries of explainability of a given epistemology [5].

2.2 Reaching an Epistemic Wall in Visualization

A recent turn in visualization research reveals that *perhaps* the limits of the foundational theories have been reached. One example of this comes from researchers demonstrating the consequences of unexamined assumptions about a typical visualization user. This line of research has exposed that visualization tools are designed with the expectation that users are data literate [135] and generally interested in engaging with new information [67]. Others have demonstrated that visualization research unintentionally overlooks certain populations, like aging adults, currently understudied in visualization research [154]. Further, Burns et al. highlight that the visualization studies focused on novices are primarily sampled from college students, resulting in a gap of knowledge about the variety of experiences novices may have when using visualizations [27]. These assumptions not only result in visualization tools that exclude certain groups of individuals but can also affect the reach of our research by limiting how we design and evaluate visualizations.

Additionally, there are several studies that investigate how individuals make sense of visualizations beyond pure perception. This research gives prominence to personal identity, prior beliefs, and personal relationship to the topic being visualized as consequential factors for users' interaction with and interpretation of visualizations [71, 77, 82, 83, 92, 123, 160, 161, 162]. In concert, recent research has examined instances where foundational visualization guidelines fall flat. In an interview with visualization authors, Hullman found that many choose not to visualize uncertainty depending on communicative goals, although guidelines advocate for it [73]. And others have explored the role of graphical embellishments that break with minimalist design guidelines, reporting that such embellishments support reader engagement [6], can contribute to visualization memorability [9, 19], and increase accessibility for individuals with intellectual and developmental disabilities [159].

Some community members have turned to other disciplines to reframe visualization problems outside of the traditional framework shaped by the foundational theories. Adar & Lee-Robbins turn to learning sciences to reformulate the relationship between a visualization designer and visualization user as one between student and teacher [1, 94]. This line of research lays the groundwork for how to use learning sciences theories to support learning and affective goals in visualization design. Others turn to behavioral economics, specifically *utility theory* as the theoretical basis for their design probe [78]. By drawing on utility theory, they justify design decisions for eliciting preferences for participatory budgeting through a graphical interface. Further, some researchers have turned to disability studies and crip technoscience to trouble the characterization of disability and offer alternative methods of designing visualizations [72, 98]. There is also a long line of visualization research exploring the productivity of introducing Bayesian statistics to understand how individuals' prior beliefs change after reading visualizations [77, 82, 83].

Others have directly critiqued foundational theories as insufficient. Kosara highlights how many results from seminal papers about perceptual studies are wrong [89]. He calls for systematic reviews to generate reliable and replicable theories so that *"visualization [can] become a real science"*. Similarly, Bertini et al. poke holes in existing perceptual theories and visualization guidelines [15]. They provocatively illustrate that if a designer strictly followed minimalist visualization guidelines that aim for perceptual precision, then they would only make scatter plots. These arguments point to the complexity of reading visualizations beyond perceptual precision and demonstrate that strict minimalist guidelines ignore affective, educational, persuasive, and other communicative goals. Further, Hullman & Gelman present the need to develop theories of how visualizations support exploratory data analysis [74] and Dimara & Perin report that the visualization community *"knows little about interaction design"* [43]. These papers, among others, point to the gap in foundational theories needed to effectively guide visualization research and design.

This dissertation argues that it is time to be more precise about how theory is discussed within visualization as a field. To this end, we identify epistemology as a more exact and productive path to discussing the recent tensions challenging foundational visualization theories. This claim does not stand alone. Others in the field have also pointed to epistemology as a central concern, identifying the myriad of issues that stem from not addressing epistemology directly.

In his survey of senior visualization researchers, Correll investigates whether the field of visualization is developing any sufficient epistemology, to the extent that knowledge occurs intentionally rather than by accident [35]. The resulting responses reflect a mixed level of satisfaction that foundational theories are relevant or even useful: *Could more explicit engagement with epistemology instill more confidence in research results? Do we even know the types of epistemological theorization that we need to move forward productively?* Losev and colleagues call for an embrace of diverse methods from other disciplines, placing responsibility on community members to actively understand and support diversity in the field [97]: *What are the actions necessary for*

the visualization community to accept research that stems from alternative epistemologies? What is the potential bridging or translation work that is required? In a project with collaborators from three distinct disciplines, Panagiotidou et al. experienced the tension of designing a visualization because the disciplinary divide meant that collaborators held starkly different epistemological interpretations of data [119]: *How would our conceptualization of data change when we approach it from an alternative epistemology? Would we be better equipped to collaborate with disciplines with different notions of data?* There is also research that highlights the epistemic misalignment between methods of inquiry and methods of evaluation — exposing the limitations of assessing research contributions without addressing epistemology within the field [34, 107]: *How do we assess the progress of the field if we do not directly engage with the underlying epistemic values that shape evaluation methods?*

One move toward addressing the epistemic wall in visualization has been the introduction of feminist theories. In the following section, we present visualization research that uses feminist and critical theories to conduct research and design visualizations from a perspective that is in line with the epistemic values of feminism and criticality. Said differently, this line of research pays attention to the ways that power and privilege shape data, visualizations, and the people who use them.

2.3 Feminist Epistemology for Visualization

The term *feminist theory* directly references scholarship that unpacks, theorizes, and relates how systems of power and privilege shape societal relations and other objects in the world. This is not to be confused with *feminism*, which broadly describes socio-political movements that advocate for equality and the demolition of systems of oppression that affect people across race, sex, ability, class, and other marginalized identities. These two feminisms are not at odds, but rather, require clarification to acknowledge the rich and varied scholarly practice of feminist theory.

On many occasions, the term *critical* will also occur next to feminism — interchanged with or used to describe feminist scholarship. While the terms are related, critical scholarship takes a broader stance than feminism, challenging existing conventions. *Critical theory* puts forward theories of how hegemonic systems impact society, history, and culture. In this dissertation, the two are separated to indicate the different origins of the ideas.

Recent visualization research has also operationalized feminist values into guidelines for design, ethics, and data. Across this scholarship, there is a shared goal to reduce differences in power, re-center context, and highlight the responsibility of the visualization researcher. To this aim, feminist and critical theories have had steady roles in shaping discourse regarding the non-neutrality and impact of visualizations. An early example comes from Dörk et al. and their characterization of *critical infovis*, which outlines a set of design guidelines that emphasize the role of the power that designers have in creating visualizations [44]. In line with this work is a variety of guidelines for researchers, designers, and analysts to use data visualizations ethically by considering the influence of power and privilege in the visualization design process [12, 33, 42, 49, 50].

Following these guidelines, there is a body of research that interprets and embodies *data feminism* principles developed by D'Ignazio & Klein [42]: *elevate emotion*, *rethink binaries*, *challenge and examine power*, *embrace plurality*, *make labor visible*, and *consider context*. In visualization design, Elli et al. elevate emotion by designing a visualization with sensitivity and care to humanize the people represented in the dataset [51]. Lan et al. further contextualize this work by cataloging visualizations that leverage emotion as a design consideration [90]. In communicating design decisions, several researchers have advocated incorporating designers' decisions as a salient component of a visualization [69, 106, 158]. In support of examining power by making labor and decision-making visible, Kauer et al. design a visual system that provides interaction modalities

for individuals to discuss together and expose otherwise hidden decisions about the design and data curation processes [80].

Along with applying feminist and critical theories to visualization processes and outputs, others challenge the foundational theories by introducing their own theoretical concepts. Hinrichs et al. challenge that visualizations represent more than engineered tools, as often characterized by foundational theories [70]. They offer the concept of *visualization sandcastles* to illustrate that visualizations represent *"ephemeral forms of [a researcher's] thinking"* that can also provoke, orient processes, and mediate discussions. Others have put forward novel theorizations about data. Offenhuber presents *autographic visualization* as a technique for dis-entangling physical representations from the data they represent [116]. While Ross et al. present another perspective on data, one that positions *data in a void* [131]. This theory challenges the idea that data are easy to access and extract. Instead, they present data as minute specks in a vast space of possible datasets that have yet to exist to draw attention to the power of the data collection decision-making process. Further complicating and challenging the foundational theories of data, is the work by Correll & Garrison that calls attention to the varied and multiple *data cultures* — *"the interconnected epistemologies, rhetorics, and technologies"* — that shaped how information about the body became data [36]. The authors illustrate how diverse sets of social, religious, and technical factors influence what is regarded as a body and how visualizations substantiate and reinforce these specific perspectives.

This prior work is critical and paves the path for applying feminist and critical theories within visualization research: highlighting overlooked areas of work, implicating research with ethical responsibilities, and introducing terms and concepts from feminist literature. And yet, only a few papers present their engagement with feminist thinking at a theoretical level, leaving an epistemological gap for the reader between what should be done and why. In other words, the value of feminist theory as a how-to guide for design principles is critical and necessary, but there is more work that theory can do.

2.4 Gap and Position

Specifically, there is more work that *epistemological theory* can do. As mentioned earlier, particular epistemological perspectives have influenced the theoretical foundations of visualization. Still, the community has not directly engaged with and discussed the merits and values that follow from pursuing a specific epistemic stance over another. This has led to implicit and specific conceptualizations of visualization. One conceptualization that falls from the foundational theories is that visualizations are data-driven and should contain little graphical embellishment to represent the data faithfully. This idea leads to another: minimalist visualization design is a predicate for optimizing the efficiency and effectiveness of visualizations to compare visualizations and determine objective rankings. And all of these assumptions rely on perceptual and cognitive theories that center on perceptual precision as the predominant factor for how people read visualizations.

This dissertation argues that an epistemic wall has been reached across various visualization lines of research. This is indicated by the influx of research turning to other disciplines for new methods to solve persistent problems. In addition to introducing novel methods from economics, statistics, and learning sciences, there are explicit requests for the community to consider building theory and embracing a diversity of methods, approaches, and values. In this emergent discourse, we call for epistemic plurality broadly and the development of epistemological theory for visualization specifically.

In this dissertation, we present *entanglement theory for visualization* as one example of creating epistemological theory for visualization (Chapter 3). This thesis also contributes case studies

that clearly articulate and demonstrate the effect of direct engagement with epistemology during the visualization research process (Chapters 4, 6, 5).

CHAPTER 3

Theory

This chapter presents a novel theory for visualization — *entanglement theory for visualization* — which transposes feminist entanglement theory [7] to visualization. The theory takes a feminist epistemological stance to reframe core concepts of visualization in relation to each other and systems of power and privilege. This chapter presents entanglement theory for visualization and discusses core concepts that are mobilized throughout the following chapters. This chapter contributes to the dissertation by presenting an epistemological theory for visualization and is based on previously published work:

> Derya Akbaba, Lauren Klein and Miriah Meyer. *Entanglements for Visualization: Changing Research Outcomes through Feminist Theory.* IEEE Transactions on Visualization and Computer Graphics (Proceedings of VIS 2024), 2025.

Backstory

The main content of this chapter draws on the theory presented in *Entanglements for Visualization.* The ideas for this work originate from online discussions with Miriah Meyer and Lauren Klein that began shortly after I started my PhD. We discussed feminist epistemology, science and technology studies, and visualization with the aim of understanding how to use feminist epistemology within a visualization research context. In our first attempt to articulate the ideas from these conversations, we submitted a paper to VIS [1] 2021 that was rightfully rejected. This rejection signaled that we had not been able to adequately or compellingly articulate feminist ideas for a visualization context. And so we turned to other projects — which turned into papers presented as case studies in this dissertation — while Miriah and I continued to familiarize ourselves with feminist theory and confront the epistemological concerns that began to materialize across our research.

Lauren, Miriah, and I returned to write the paper three years later. This paper was accepted and received the Best Paper Award. The differences between the papers are twofold. The first is the

[1] VIS stands for IEEE VIS, a conference and top venue for publishing visualization research. IEEE is the organization (Institute of Electrical and Electronics Engineers). VIS is the conference name (**V**isual Analytics Science and Technology (VAST), **I**nformation Visualization, and **S**cientific Visualization).

inclusion of a concrete case study — presented in Chapter 5 — that demonstrates how engaging with feminist epistemology changed our research outcomes. The second is a more precise and succinct articulation of the epistemological theory, which is presented in this chapter. Crucial to this contribution is the *transposition* of theoretical concepts from feminist epistemology to visualization. As one of the core contributions of this thesis, it seeks to outline and demonstrate what epistemological theory for visualization can look like. This chapter concludes with a set of core concepts from feminist epistemology that are mobilized across the following chapters and my PhD work.

3.1 Feminist Epistemology: Entanglement Theory

Entanglement theory is a theory of knowledge that centers relationships as the source of meaning. Broadly, entanglements is concerned with how knowledge is derived from and made meaningful from relationships between humans and non-humans, objects and processes, and individuals and systems. The theory has seen recent uptake within HCI [41, 59, 91, 133, 143, 153], but until our work, it had been missing within visualization. More specifically, our work draws on the theory of Karen Barad, presented in their seminal text *Meeting the Universe Halfway: Quantum Physics and the Entanglement of Matter and Meaning* (2007) [7]. Building on Barad's theory within a broader genealogy of feminist scholarship, we present *entanglement theory for visualization*: a transposition of core epistemological concepts from feminist theory onto visualization objects of study.

In the following section, we review the genealogy of entanglement theory. A genealogy as opposed to a history, does not claim an objective historical recounting [52, 55]. We employ a genealogy to intentionally situate entanglement theory within the historical context of feminist ideas, people, and theories.

3.1.1 Genealogy of Entanglement

Our genealogy begins with the *science wars*: a period in 1990s USA marked by debates between scholars about the axiology and epistemology of science at the time. Specifically, postmodern theorists critiqued values around objectivity and neutrality, drawing attention to the many ways socio-technical influences like sexism, racism, political trends, and economic priorities have shaped science. Underlying postmodernism was the epistemology of *relativism*, which rejects the possibility of a singular objectivity and instead claims that all knowledge is relative to specific individuals. This assertion gave rise to heated debates, where scientists accused postmodernist scholars of being anti-intellectual and ignorant about scientific methods.

American feminist scholars contributed to this discussion, introducing an alternative epistemology. An important figure in this history is Donna Haraway, who spoke of the *god trick* in science [63]. The god trick describes how visual methods in science use the view from above when representing data, scientific results, and other outcomes. This perspective instills the idea that it is possible to see *"everything from nowhere"*, perpetuating a belief that knowledge is created by no one and nowhere specifically [63]. Lacking a position, the god trick successfully cloaks the influence of politics, capitalism, and patriarchy on knowledge claims. When positionality is invisible, it becomes difficult to challenge.

As an alternative, Haraway presents feminist objectivity, or *situated knowledges* [63]. Unlike relativism, the theory asserts that truth can exist, but that individuals can only gain knowledge through partial and situated perspectives. More specifically, individuals' unique perspectives

shape how they know what they know, situating the person and their knowledge in a specific time, place, and context. The theory continues that truth, or feminist objectivity, rests on composite situated knowledges, a blending and complementing of multiple partial perspectives. This theory has led other scholars to apply situated knowledges across a wide range of disciplines, unpacking the specifics of the constituent partial perspectives and how these situated knowledges contribute to a more comprehensive perspective of some phenomenon in the world [28, 130, 140].

Following Haraway, many scholars have developed theories that expand and complicate the nature of knowledge from a feminist perspective. One such theorist is Karen Barad, who developed the theory that we build on in this chapter. Before turning to feminist philosophy of science, they were a theoretical physicist. It was in their engagement with the limits of traditional Newtonian physics that they began to build their theory. Published in *Meeting the Universe Halfway: Quantum Physics and the Entanglement of Matter and Meaning* (2007) they grappled with the inseparability of people, scientific instruments, and other social, historical, and political influences from scientific knowledge. Entanglement theory, or agential realism, lays the groundwork for a feminist epistemology that ties together ontology, ethics, and the *"human and nonhuman, material and discursive, and natural and cultural factors"* that constitute knowledge [7, p.26].

Barad's theory has played a critical role in the subsequent feminist and science and technology scholarship. Contemporary feminist scholars are building on entanglements and feminist ethics from a variety of intersectional perspectives like queer theory, post-humanist theory, Black feminism, and Indigenous studies. Across this scholarship, the emphasis lies on understanding how relationships create and shape knowledge. Together, the multiplicity presented across this body of feminist scholarship presents the many and situated ways that knowledge is produced. Additionally, it provides an analytical lens with which to understand how these relationships are shaped by systems of power — racism, sexism, colonialism, ableism, capitalism, and others — and then, in turn, shape how we come to know things about the world.

3.1.2 Entanglement Theory

We were drawn to engage with Barad's theory due to their theorization of uncertainty, an overlapping concern of visualization research and quantum physics. Barad begins their articulation of entanglement with competing interpretations of a thought experiment in quantum mechanics. The thought experiment examines the trade-off between measuring the momentum of an electron and knowing its location. Here lies the paradox: both position and momentum cannot be known simultaneously.

One explanation of this paradox comes from Werner Heisenberg and is known as the *uncertainty principle*. The principle asserts the inverse relationship between the certainty of knowing an electron's position and its momentum. In other words, knowing more about one attribute increases uncertainty in the other. Barad presents another interpretation of the thought experiment, one offered by Niels Bohr to describe the paradox. In their retelling, they suggest that Bohr questioned the existence of an electron's position or momentum before it was measured. Rather, these properties came to exist only as an artifact of interacting with the measurement device. In this interpretation, properties of the electron are *indeterminate* rather than uncertain.

Building on the ideas from this thought experiment, Barad develops an epistemological theory that also describes how objects come to be (ontology) and our responsibility in their production and study (ethics). This theory of entanglements, or *agential realism* as Barad calls it, posits that the act of studying something in the world creates both knowledge about the phenomenon and also the attributes of the phenomenon. Knowledge-making practices *make* properties matter. As a result, choosing to study one thing over another ties knowledge to specific and situated research practices and priorities. Tightly coupling ethical reflexivity with scientific practice sits

at the core of feminist commitments to examine how power and privilege impact how we come to know the world.

Entanglement theory begins with a *phenomenon* in the world as the object of study. Phenomena do not exist as discrete objects but instead are entangled with dynamic relationships: to humans, non-humans, history, society, technology, processes, and power, among others. In order to study a phenomenon, we must use some *apparatus*. This apparatus may be a measuring device or a process, such as data collection and analysis methods. Apparatuses shape phenomena — determining what is both knowable and merits knowing. This definition of apparatus extends to systems of oppression as powerful determinants of knowledge. Further, apparatuses, like phenomena, are dynamic and entangled.

When an apparatus and phenomenon interact, a static and bounded representation is produced. This representation — which we call a *knowledge artifact* — is the object, process, and idea that can be known and studied. Crucially, although the representation is itself static, it encapsulates all the entanglements of the phenomenon and the apparatus. Entanglement theory outlines an epistemology that centers the relationships that constitute phenomena and the apparatuses as critical components of any knowledge claim.

3.2 Transposing Theory: Entanglement Theory for Visualization

We transpose the key concepts from entanglement theory onto a visualization context, focusing on data, visualization, and insight. *An entanglement theory for visualization reinterprets data, visualization, and insight as knowledge artifacts, produced through the interactions of phenomena and apparatuses* [5]. With this epistemological reframing, we bring to focus the complexity of relationships that inform and shape visualization research objects of study.

3.2.1 Entangled Data

Data are knowledge artifacts created by apparatuses like measurement devices and models to represent some phenomenon in the world. While the data themselves are static, they represent phenomena that are dynamic and always changing. The apparatuses that give shape to data include processes, like data collection standards and conventions, and measuring devices. Data are entangled with the positionalities and motivations of the individuals who design, collect, and verify them [76] — reflecting what phenomena are considered countable and not, at what point in time, and how phenomena are measured. Some of these entanglements are recorded as metadata [26] and others as uncertainty metrics [117], but entanglement theory calls for a broader consideration of the relationships that constitute data-making apparatuses.

Entanglements draws our attention to the numerous influences that stem from structurally oppressive systems. Census data relies on cultural and time-specific demographic categories to silence some groups over others [20]. Natural language models are trained on news articles that reflect cultural gender bias and thus create sexist associations in algorithms, like *woman is to nurse as man is to doctor* [17]. Commercial facial recognition software perpetuates bias, performing worse on identifying Black women's faces, which are data points notably missing in the datasets [25]. Importantly, entanglements draws our attention to the ways that these technologies perpetuate bias across intersecting and marginalized identities.

These are some examples of how data as knowledge artifacts are shaped by and partially representative of the dynamic and entangled apparatuses and phenomena. Entangled data exposes

additional considerations for the visualization research community, like how to incorporate ethical responsibility and accountability when working with datasets: *What is our responsibility to account for, understand, and take responsibility for the entanglements of a dataset?* Further, it challenges how we design tools for data analysis: *How do our design practices change with an entangled data perspective? Should we visualize entanglements?* Entangled data shakes the ontological assumptions of visualizations as objective representations of the world, compelling us to consider: *What does visual data analysis represent if data are subjective, entangled, and partial?*

3.2.2 Entangled Visualization

Like data, we present visualizations as knowledge artifacts created and shaped by a designer's formulation of a specific visualization problem. The formulation can come in many forms and are widely discussed and familiar to the visualization research community. As an example, a visualization can include a visualization researcher's interpretation and operationalization of a domain expert's data analysis needs [137]. It could also include a designer's objective to share personal data [100] or emotional stories [51]. Across all these formulations and motivations, the visualization is only a momentary snapshot of the phenomenon being visualized.

From the perspective of entanglements, visualizations can only represent a static view of phenomena that are entangled, dynamic, and evolving: research questions change [4] or analysis needs develop [156]. Similarly, the apparatuses used to create the visualization are also entangled and complex. Such apparatuses include a designer's formulation of data analysis task and decisions on how to visually represent the data [107]; conventions that inform design and interaction choices [15, 68]; and the availability of tech stacks [128]; among many others.

The entanglements of apparatuses point to the complexity of visually representing the world *and* the power structures that shape what and how phenomena are visualized. Visualizations are not neutral [33, 44, 46, 107] and perpetuate a specific way to visualize data. These graphical conventions are drawn from statistics and are intended to imbue a rational and neutral view from above [68]. These conventions are entangled with oppressive histories that have rejected designerly ways of knowing [3] and the work of marginalized individuals [11, 85] as unserious and art-like, consequently excluded from visualization history.

Entangled visualizations ask us to reflect on our assumptions of how we value and evaluate visualizations: *What is a good visualization and who decides? What value systems do these judgments sustain?* If visualizations represent entangled, dynamic, and changing phenomena, it bears to question the point of having a visualization system to begin with: *What is the value of creating visualization tools if they can no longer represent the phenomenon?* Ultimately, entangled visualization is a call to seek diverse perspectives to understand the role and impact of visualizations.

3.2.3 Entangled Insight

The oft-stated goal of designing visualizations is to support insight by putting graphical form to data in order to leverage cognition [29, 57]. Insights are knowledge artifacts entangled with the people who generate insights and the entanglements of the data and the visualization. Stated differently, insights are emergent and inseparable from data practices, a visualization user's situated knowledge, visualization conventions, available technology, and all the entanglements that shaped the data and the visualization tool.

Across studies, there is an abundance of research exploring and unpacking the entanglements between people, data, and visualizations. This work reveals the importance of personal beliefs

on interpretation [92, 161] and on trust [81, 123, 162]. Other research has demonstrated the role of domain knowledge and experience in insight generation [45]. These are but some of the entanglements that shape and create insights.

Reframing insights as a knowledge artifact results in a proliferation of entanglements, leading us to consider how people, visualizations, data, history, privilege, society, technology, power, and more shape how insights come to be. This shift in perspectives challenges and makes it more challenging to appraise the role of visualization tools in generating insight: *How do we assess the effectiveness of a visualization if it is one of many factors that contribute to the development of an insight?* Further, it complicates our own role in future knowledge generation: *What are our ethical responsibilities when we choose to create tools for certain disciplines over others?* This question leads to a broader discussion for the field: *How do we assess the impact of our work specifically when it is entangled with a bunch of other stuff?*

3.2.4 Entangled Ethics

By interrogating the common knowledge artifacts of visualization — data, visualization, and insight — we are left with ethical questions on how to conduct our research. Although ethics is not a concern new to technological development, entanglements casts light on the shortcomings of ethical guidelines that fail to consider the complexity and dynamic nature of technological development, use, and responsibility. Current ethical guidelines are checklists, often developed without considering entanglements in mind. These guidelines can lead to *moral stress* when individuals take on the personal responsibility for upholding the ethical recommendations through daily actions that may or may not have broader organizational support [127]. Or these guidelines can be hopelessly inadequate if we consider the sheer amount of complex actors that contribute to developing and shaping a dataset [93] or the range of competing interests and autonomy that lie at the crossroads of collaborative decision-making [4].

Entangled ethics makes clear that ethical frameworks developed *ad-hoc* or *post-hoc* are inadequate to deal with such complexity and dynamism. Further, entanglement theory places ethics central to visualization research processes, inseparable from the decisions and entanglements that make up the work. From an entanglements perspective, creating knowledge is an ethical exercise: *What topics receive our attention? What topics, conversely, are ignored?* Further, it brings forth different systems of privilege that support certain research endeavors over others: *What funding opportunities exist and for what types of research? Who has the opportunity to conduct research?* Entanglements are a reminder that historical influences shape knowledge and, similarly, will shape what comes after.

Critically, entangled ethics shifts the ownership of ethics from an individual researcher within the bounded context of a single research project to the set of relationships that are the foundation of and foundational to collective ethical accountability. Generating knowledge is a collaborative endeavor, and those choices will perpetuate (or suppress) some knowledge over others. Engaging in entangled ethics requires constant reflexivity to question the decisions behind research decisions.

3.3 Mobilizing Feminist Epistemology for Visualization

This dissertation reflects the countless hours spent engaging with feminist theories, discussing the epistemological implications, and transposing the core ideas to a visualization context. While we draw heavily on entanglement theory, as articulated by Barad and explained in the previous sections, there is a significant amount of additional interpretation that we have conducted in

order to mobilize feminist epistemology in the practice of designing, developing, and studying visualizations. This section further articulates the set of epistemic and ethical considerations that have guided the work presented in this dissertation. These interpretations extend and clarify entanglement theory for visualization and are referenced across the thesis, drawing the case studies together.

As an overview, the mobilized concepts are as follows:

- **Entanglements beget investigation.** Science is shaped by the material, social, and historical world, and understanding these relationships enriches knowledge contributions.
- **Situated knowledges requires commitment.** Claims to knowledge are always partial and situated in the circumstances from which they arose. Such partiality gives rise to accountability to understand the limits of knowledge claims.
- **Ethics are not an afterthought.** Ethics are inseparable from how science is conducted, as any decision to study one thing is also a decision to ignore another.

Entanglements beget investigation. Entanglement theory for visualization lays the groundwork for unpacking everyday objects of visualization research in order to *implode* [48] the many people, places, and histories that shape it. This perspective inevitably leads to pulling on different types of threads to understand how and why things come to be in the world. In Chapter 4, we explore why visualizations follow minimalist design guidelines. We demonstrate how entanglements drew our attention to the tensions present in studying and using the term *chartjunk*, revealing hidden entanglements tied up in an epistemic and ideological dispute. Entanglement theory leads to developing a sensitivity to question the unquestioned, recognizing that some assumptions are invisible by design and reflect larger hegemonic systems.

Situated knowledge requires accountability. Falling out of investigating entanglements is how feminist epistemology theorizes about knowledge. Knowledge, as a product of dynamic and specific entanglements, is situated, partial, and imperfect [63]. Crucially, from this perspective, knowledge is not neutral and a product of systems of power and privilege. Therefore, situatedness requires taking accountability for the ways we are entangled with and entangle the knowledge we develop. In Chapter 5, we work with this concept to realign data and domain expertise as two situated knowledge sources about a phenomenon in the world. In practice, this manifests as developing alternative visual encodings that utilize the strength of visualization interaction and also convey situatedness. Viewing data and knowledge as different sources of situated knowledge — shaped by and entangled with many other influences — lends an alternative perspective that challenges what belongs in an analytical visualization tool. Thus, by designing with a sensitivity to situated knowledges, we take responsibility for how we treat data and knowledge in visualization design.

Ethics are not an afterthought. The final mobilized concept regards ethics. Ethics in entanglement epistemology is embedded in and inseparable from all research processes. Feminist care ethics embraces that ethics is emergent, relational, and situated [40]. Neglect and power are ever present in knowledge creation practices: the choice to study one topic is entangled with the choice to ignore another. This compromise emphasizes that enacting care within research is a direct result of power and privilege. And so, care ethics are entangled with every research decision and process. This particular sensitivity brings to focus what and who lack care. In Chapter 6, we bring forward the entanglements in visualization research collaborations, drawing attention to where our process models neglect care. The work in this chapter draws on personal experiences conducting collaborative visualization research with domain experts, where the visualization tool ended up as the least interesting part of the research.

CHAPTER 4

Entanglements in Design Guidelines

In this chapter, we expose how the entanglements of a standard design guideline — *avoid chartjunk* — carry rhetorical weight that is both harmful and unproductive for visualization research. To this aim, we present the political, personal, and epistemological motivations of chartjunk. It is supported by the following secondary contributions: the etymology of chartjunk and artifacts from performative acts intended to help the visualization community grapple with the entanglements of chartjunk across our research. By exposing the entanglements of chartjunk and performatively taking out the trash, we take a step toward removing chartjunk from the design guideline lexicon. This chapter is based on the previously published work:

> Derya Akbaba, Jack Wilburn, Main T. Nance, Miriah Meyer. *Manifesto for Putting 'Chartjunk' in the Trash 2021!* alt.VIS, a workshop co-located with IEEE VIS, 2021.
>
> Supplemental Material:
> https://jackwilb.github.io/chart-junk/

Backstory

The work presented in this chapter comes from a project between me, Jack Wilburn, and Miriah Meyer. It began with a reading group discussion about a recently published paper [159] that presented a study about the benefits of chartjunk for individuals with intellectual and developmental disabilities. During that discussion, we noticed the tension in the paper between the use of the word chartjunk — *-junk* with connotations of trash and uselessness — and the many benefits discussed by the authors. Our work with feminist scholarship drew our attention to this tension and to question *why* the word chartjunk existed in the visualization community's lexicon in the first place.

Coincidentally, IEEE VIS hosted the inaugural alt.vis workshop that year. Such a venue called for *alt*ernative thinking and gave us the space to experiment with performance art as a communicative device. The performance art was inspired by an exhibit I had seen on Mierle Laderman Ukeles at the Queens Museum in New York, USA (c.2017), and had stuck with me due to its critique on gender, class, and race in maintenance roles. Together, the work presented in this chapter

draws attention to chartjunk as a rhetorical device within visualization guidelines that needs contextualization, interrogation, and a little maintenance.

4.1 Case Study: Revealing Rhetoric

Chartjunk is a common word used in the visualization literature to describe a class of graphical elements like redundant data encodings, flourishes, icons, and other imagery [9, 19, 121, 159]. Originating from Edward Tufte's self-published book [145], the word is a rhetorical device used to group and categorize what he described as unnecessary and redundant elements in a visualization. Following other foundational theories that strove for perceptual precision and statistical accuracy, Tufte reinforced the convention that visualizations should uphold minimalist standards in design, simultaneously discouraging alternative approaches. To more completely understand the gravity of chartjunk within the visualization community, we explored the history, or etymology, of the term, revealing that beyond rhetorical, chartjunk also represented an ideological divide.

4.2 The Entanglements of Chartjunk

Establishing the etymology of chartjunk follows feminist theory: a recognition that knowledge is derivative of and entangled with objects in their contexts. In this instance, the object under investigation is the word chartjunk. Chartjunk was defined in Tufte's 1983 book *The Visual Display of Quantitative Information* [145]. The book summarizes and presents his standards for good visualization design. The principles are informed by his training as a statistician and political scientist. In part from this epistemic position, he asserts that visualizations should *"reveal"* and *"tell the truth"* about data through minimalist design choices — reinforcing the neutrality of science and data.

Tufte uses illustrations from a contemporary designer and the first visualization designer at *Time Magazine*, Nigel Holmes, as examples of bad chart design. Pointedly and abrasively, Tufte say that Holmes creates visualizations that *"lie"* and *"lack statistical integrity"* [145, p79]. He justifies these claims by linking Holmes' training as a designer instead of a statistician. Such severe critiques continue in Tufte's second book [144], calling Holmes' work *"unsavory"*, *"chockablock with cliché"*, and lacking credibility on account of chartjunk [144, p34–35].

In response to Tufte's critiques, Holmes counters that Tufte's resolution to minimalist statistical charts is naive and ignores other use cases of visualization, like grabbing attention and communicating to an audience [122]. This debate between the two men positioned statistical and designerly approaches to visualization design as diametric and opposed.

Within this context, Tufte outlined design guidelines to communicate his standards of rigor for graphical excellence. One of these standards includes eliminating graphical elements and ornamentation that do not exclusively represent data, which he provocatively named *chartjunk*. He made his position clear: *"graphical decoration"* is not scientific and has no role in a data visualization [145, p76, 107]. His minimalist guidelines even go so far as to remove grid lines and axes.

Beyond etymology, which exposes the history of an object, entanglements are a reminder that origin stories are inseparable from present-day actions and reactions. More specifically, the visualization community's engagement with and uptake of chartjunk is another crucial aspect of understanding chartjunk's entanglement with visualization research. By using chartjunk as an academic concept that is taught and studied, the visualization community has reinforced a specific epistemological stance on design. In other words, the design guidelines reinforce the

epistemological stance that data are neutral and apolitical and *also* discredit ways of knowing that stem from different disciplines like design.

Tufte's self-published books received high acclaim, and his guidelines were readily adopted into visualization's vocabulary and design guidelines. Academics and practitioners alike employed chartjunk in various ways, becoming a general word to describe a variety of graphical components with negative connotations. Despite chartjunk's uptake, it was not without complaints. Some have warned against chartjunk's overarching negative connotation, suggesting that it could unintentionally describe otherwise useful graphical elements [53, 88]. While others have noted the inherent tension in calling useful graphical elements *junk* [88, 121]. As an intermediate solution, chartjunk is used with other more descriptive words: *"embellishment [9, 159], excessive annotation [19], decoration [19], extraneous elements [75], and unnecessary ornamentation [75]."* [3]. Further, in an interview study with practitioners, Parson & Shukla found that chartjunk served as a stand-in for bad design — bad design varying across individuals and reflecting *"underlying commitments to a particular design philosophy"* [121]. These examples reveal an unnecessary definitional and epistemological gap created by the term chartjunk. The word encapsulates too much and is subjectively interpreted due to its negative connotation.

The matter is further complicated with several empirical studies illustrating the benefits of graphical elements that could be described as chartjunk. Across different studies, research has shown that visualizations containing chartjunk are more memorable in certain scenarios [9, 19], can increase engagement and exploration of news media [6], and can support individuals with intellectual disabilities [159]. These studies and the necessity for more nuance demonstrate that chartjunk is unproductive in some settings.

This is the context with which our work engages. Presenting chartjunk's complicated history as a provocation, we turned to the visualization community and asked: *Where do we go from here? Do we continue to fumble for words that attempt to neutralize the negative connotations of junk? Do we continue to devalue designerly and other non-minimalist perspectives of good visualization design? Do we continue to produce incongruities in how we write about, talk about, and think about what makes visualizations effective?* [3].

4.3 Revealing the Entanglements of Chartjunk and Visualization Research

As a first step toward dissolving chartjunk from the visualization lexicon, we employed feminist maintenance art to provoke and illustrate the widespread entanglements of chartjunk within visualization research. The performance, consisting of three parts, is a metaphorical act of taking out the trash. The first act is *the edit* of the Wikipedia page about chartjunk to include the entangled history and etymology of the term. The second act is *the video* capturing the performance of cutting out the word chartjunk from papers that use the word in the IEEE library. And the final act is *the scanned repository*, a set of scanned papers with empty spaces where chartjunk used to be — an invitation to reimagine words that may fit better.

The performances are an homage to Mierle Laderman Ukeles, who performed acts of maintenance as art and activism in New York City, USA, in the 1970s. By reframing maintenance as art, her performances drew attention to acts of gendered, classed, and invisible labor required to maintain families, cities, and society. Inspired by her work, we hope our maintenance art reframes how the visualization community thinks about charjunk.

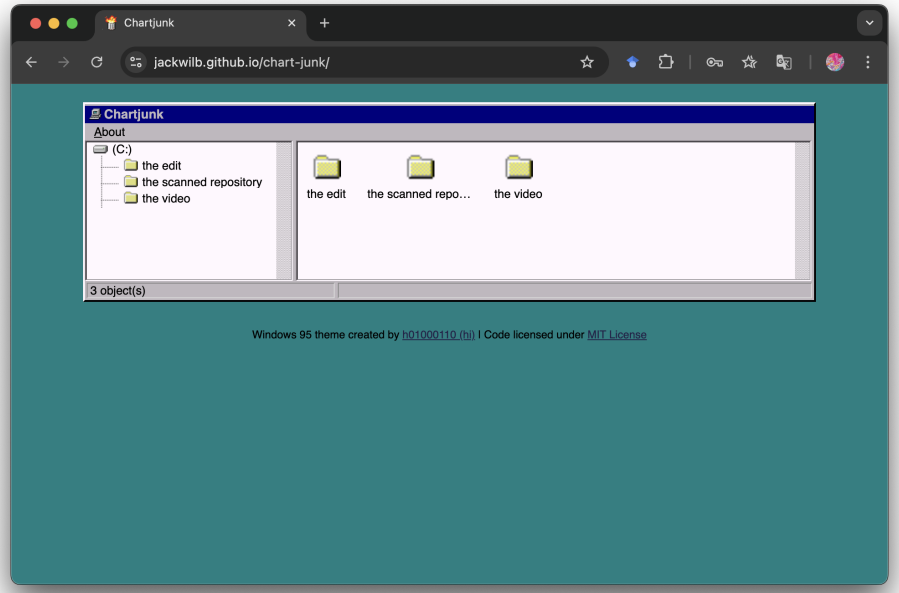


Figure 4.1 — The website holding the artifacts from performing maintenance art. Three folders represent the three different performances for revealing the entanglements of chartjunk with visualization research.

Each part of our performance art is stored virtually [1]: a triptych of digital traces to serve as reminders that maintenance requires constant revisiting and attention.

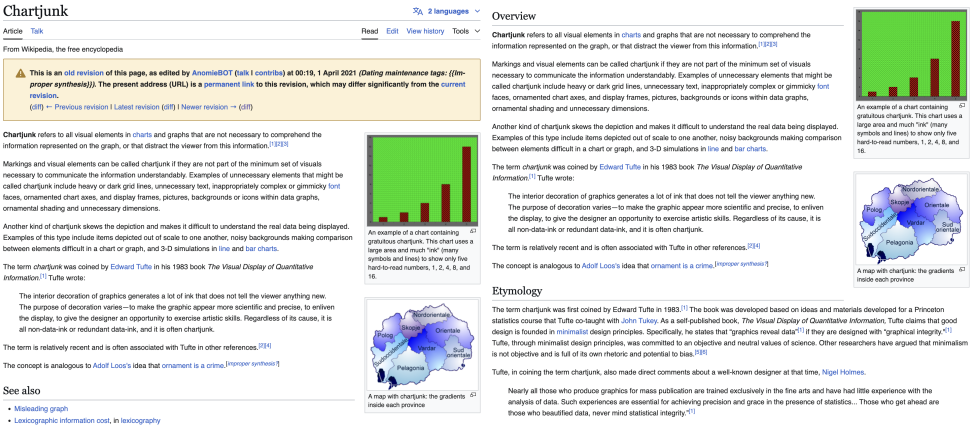
The first act *acknowledges* the entangled history of chartjunk and brings it to a public arena. To accomplish this, we edited the English-language Wikipedia page about chartjunk [2]. In our edit, we included the relevant historical and social context of chartjunk, adding a new section to the page about the term's etymology. We also added a section about the benefits of using chartjunk reported across different visualization studies. We intended to provide context and to complicate a monolithic narrative that chartjunk in visualization design is bad.

4.3.1 The Video

The second act *removes* chartjunk from the visualization lexicon by cutting printed research papers. We printed out a corpus of papers from IEEE Xplore — the digital library that holds the VIS conference proceedings — which were published at the time and contained the word chartjunk.

[1] The artifacts were stored initially on chartjunk.art but unique domain names require maintenance in the form of funding. So now the artifacts are freely hosted on GitHub: `https://jackwilb.github.io/chart-junk/`.

[2] `https://en.wikipedia.org/wiki/Chartjunk`



(a) Before The Edit. (b) After The Edit.

Figure 4.2 — A comparison of the Wikipedia page on chartjunk. One big difference is that it includes a section about etymology. A full edit history can be found online.



Figure 4.3 — Still from *The Video*. In this scene I am cutting out the word chartjunk from a printed copy of a visualization paper.

As part of the performance, I sat down in a beige top in a seminar room and recorded myself (with the help of Ed Cask) cutting out tiny squares of paper where chartjunk used to be. The result:

a boring video. The boringness mimicked Ukeles' sentiment that maintenance is unglamorous, boring, and tedious: *"Maintenance is a drag; it takes all the fucking time (lit.)"* [147] We wanted to convey this sentiment to the visualization community to spark discussions around maintenance: *where and with whom does responsibility for maintenance lie?*

4.3.2 The Scanned Repository

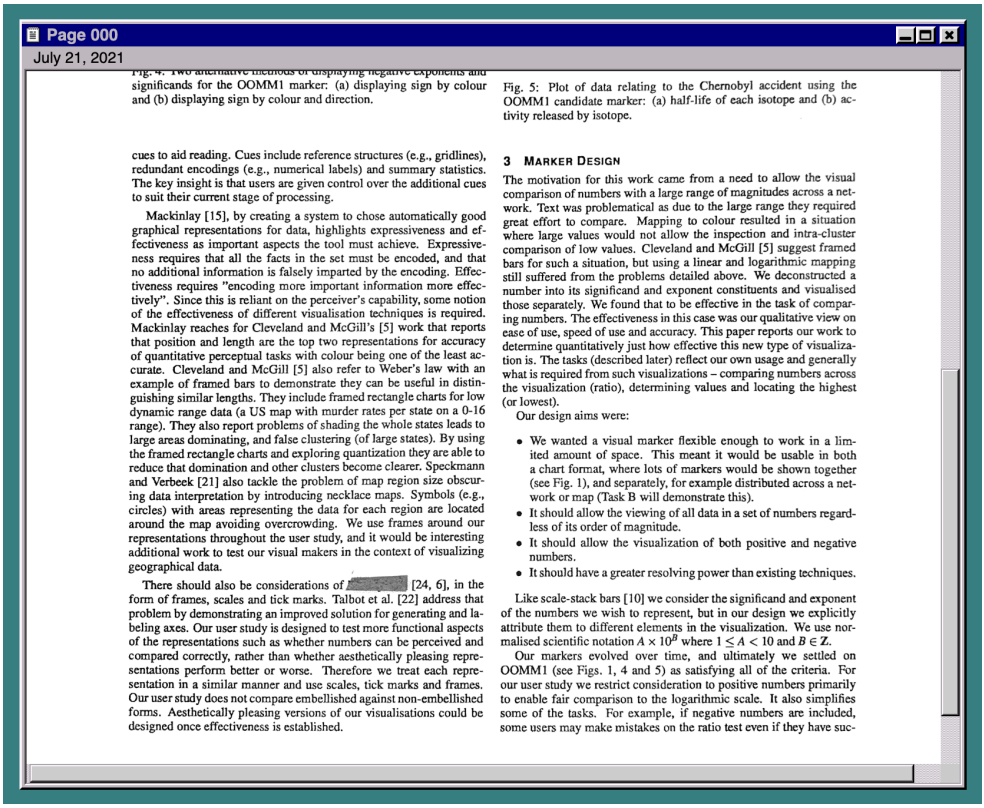


Figure 4.4 — Page 000 from *The Scanned Repository*. In the top right corner there is a blank in the sentence: *"There should also be considerations of [——], in the form of frames, scales, and tick marks"*. We ask: what may fit better here?

The last act of the tritypch invites *re-imagination* from the visualization community. It is a scanned repository of the papers from the video. The papers have blank spaces where the word chartjunk has been cut out, leaving a MadLibs-type fill-in-the blank research paper. The empty spaces are opportunities to consider what other words may better fill these spots. The pages are separated from the rest of the research paper to convey that our maintenance work is not targeting specific researchers but only the concept itself.

When the paper was originally published, we encouraged others to send their suggestions of alternative words to our fictitious co-author, Main T. Nance. We created a social media handle for Nance on Twitter (@SayNo2Chartjunk) and encouraged posts that would link the scanned

repository page from the website to a pre-generated Twitter message template. In this way, we opened the doors for collective maintenance.

4.4 Reflections for Dissertation

> "After the revolution, who's going to pick up the garbage on Monday morning?"
> - *Mierle L. Ukeles* [147]

Understanding how concepts are shaped by and likewise constitute the context in which they are derived is an important act of entanglements epistemology that has implications for how visualization research is conducted. This particular sensitivity drew our attention to the tensions present in the study by Wu and colleagues [159]. We wondered *why we, the visualization community, used the term chartjunk to describe positive visualization experiences?* In response, we investigated the unquestioned history of chartjunk as a design guideline with the recognition that objects are entangled in their historical, political, social, and material dimensions [7] and the understanding that these entanglements can change what we know about the object [48].

Focusing on the etymology of chartjunk revealed the rhetorical power of the concept *beyond* its basic definition. We came to understand that the genesis of chartjunk was shaped by politically- and epistemologically-pointed intentions, specifically against designerly ways of knowing. Tufte's critiques were informed by his training as a statistician and political scientist, and consequently, he supported minimalist design standards in his work. This stance placed other types of design as incompatible with his standards for design and visualization. When visualization researchers and practitioners began using chartjunk in their work, they inadvertently gave credence to one design practice over another.

Aside from the political and ideological implications, embracing entanglements can lead to new ways to design visualizations. In a study by Alebri and colleagues, the authors explored the relationship between chartjunk in news media with reader engagement [6]. The study demonstrates how different aspects of chartjunk — like the saliency of the embellishment or the stylistic choices — affect sense-making, relatability, and trustworthiness of the visualization. The authors advocate for a more nuanced typology of chartjunk that goes beyond a binary classification of visualizations as embellished or not, to more effectively design visualizations for public consumption.

In other work that also investigates the entanglements of a common visualization concept, Burns et al. reveal the research tensions that arise from using the concept *novice* in theory versus in practice [27]. Their study demonstrates that novices theoretically mean individuals who lack certain visualization skills, but in reality tend to represent a limited participant pool, like college students. The consequence of the study is alarming as it raises questions about the applicability of visualizations for novices. It points future research efforts toward more precisely aligning study design with design goals.

Across these examples, we demonstrate how concepts' entanglements may block productive discourse and hinder research goals. As a corrective, we can actively embrace the entanglements of the phenomena we seek to learn more about — data, visualization, and people — to move past unproductive, imprecise, and misaligned concepts.

CHAPTER 5

Situating Knowledge in Visualization

In this chapter, we describe how we developed novel visualization design guidelines by using feminist theory to reframe design constraints about data and knowledge. More specifically, our engagements with feminist theory drew our attention to recurring tensions when designing collaborative visualization analysis tools for domain experts. These tensions resulted from foundational guidelines that gave precedence to data as a first-class object over our collaborators' knowledge when designing visual systems. To overcome this tension, we recharacterized the problem through feminist epistemology, which led to a new understanding of data and knowledge as co-valued inputs of visualization systems. The operationalization of feminist theory in this way not only bridged the gap between foundational visualization guidelines and abstract feminist epistemological theory but also opened up novel design spaces in collaborative visualization tools. This chapter rests upon the work presented in the following papers:

> Haihan Lin, Derya Akbaba, Miriah Meyer and Alexander Lex. *Data Hunches: Incorporating Personal Knowledge into Visualizations.* IEEE Transactions on Visualization and Computer Graphics (Proceedings of VIS 2022), 2023.
>
> Supplemental Material:
> https://osf.io/syjkf
>
> Derya Akbaba, Lauren Klein and Miriah Meyer. *Entanglements for Visualization: Changing Research Outcomes through Feminist Theory.* IEEE Transactions on Visualization and Computer Graphics (Proceedings of VIS 2024), 2025.

Backstory

The work presented in this chapter is the result of two papers published two years apart and reflects the journey of learning how to communicate feminist theory to the visualization community. In *Data Hunches*, we mention feminist theory only briefly and as a small part of the discussion section to avoid distracting the reader from the conceptualization and operationalization of data hunches. In *Entanglements for Visualization* — after having more time to reflect and engage with feminist epistemologies — we lead with feminist epistemological theory and use the work in *Data*

Hunches as part of a case study to illustrate how feminist epistemology changed our research outcomes. In this backstory, I merge these two experiences together to provide a contextualized overview of the core ideas in this chapter.

The ideas presented in this chapter begin with the research of Miriah Meyer and one of her former students, Nina McCurdy, who published the results of their collaboration with public health experts [104]. The main result of this collaboration was not a new visualization technique, but rather a new concept: *implicit error*. Nina and Miriah developed this concept in response to something odd that they experienced in their collaboration — despite the fact that Nina had developed a visualization following standard visualization guidelines and using official data sources, the experts had to *" 'suspend [their] disbelief' around the quality, consistency, and availability of the data"* in order to use the tool [103]. Through further investigation and discussion, they developed the term implicit error to describe the relationship between data, the phenomenon it is intended to describe, and expert knowledge. While this conceptualization acknowledged the knowledge of experts, ultimately, it did not resolve the tension that data are *"official and authoritative but also deeply flawed, while the expert knowledge was partial and subjective but rich with nuance, context, and insight"* [5].

Miriah — co-leading a lab with Alex Lex at the time — had ongoing discussions about the tensions of visualizing data that were flawed and partial representations of reality, while seemingly ignoring the rich expert knowledge required to make sense of the visualized data in relation to the phenomenon it represented in the world. This tension kept reoccurring across design studies led by their students with surgeons [95], meteorologists [126], and families [108, 109]. I similarly experienced this in an early design study working with neuroscientists. My collaborators had extensive network data about the neural connections in retinas, but when this data was visualized, it more faithfully represented annotation practices instead of the underlying biology. These collective experiences challenged foundational theories that data were context-independent and accurate representations of reality, troubling foundational visualization guidelines.

As a group — Haihan Lin, Alex, Miriah, and I — returned to the conceptualization of implicit error. Through reflective practices [54, 136], we realized that expert knowledge was rarely recorded or present within the datasets we worked with and designed from. We iteratively developed and refined our individual and collective understandings through design sketches, engagement with theory, and prototype development. Haihan and I worked together to create a series of low-fidelity sketches exploring the design space of recording expert knowledge about visualization [1]. These sketches sparked feedback from the rest of the team and helped us to further refine both the concept and the graphical expressions.

As we worked on the visual concepts, we also discussed the work in relation to feminist scholarship and critical data studies. Miriah and I brought forward alternative formulations of data from critical data studies and feminist epistemologies that challenged our normative-aligned assumptions that data could be objective and ahistorical. These critiques highlighted that data are non-neutral artifacts of decision-making processes, reflecting the entangled contexts that create them [7, 42, 46]. This perspective made space for experts' knowledge as valuable inputs in visualization systems because often, the expert had extensive knowledge of the data's entanglements. From a feminist theoretical perspective, data and expert knowledge were both situated knowledges [63], co-valued and indispensable in reaching objective sensemaking with a visualization. By reconceptualizing data as another situated knowledge source, we changed how we viewed the design hurdle presented by implicit error. More specifically, it became clear that data and expert knowledge about a phenomenon were different, but equally valuable perspectives that could contribute to a more complete visualization if displayed together. Along with this epistemological shift, we reframed

[1] These early iterations can be seen in our supplemental material `https://osf.io/bqkgt`.

implicit error as *data hunches*, shifting the emphasis from what the data lacked to what the experts knew.

The final part of the work required bridging the theory with actionable visualization guidelines. In this part of the project, as Haihan developed a prototype [2], it became clear that it was not enough to develop feminist theory for visualization, but rather we also need to operationalize the theory into visualization concepts like data types and visual idioms. The affordances of the prototype deepened our ability to communicate and ground the theory of data hunches in concepts familiar to visualization scholarship.

5.1 Case Study: Impacting Design

Across many disciplines, data are used to make and justify decisions [24, 56, 142]. Crucially, those who work with data are also aware that datasets come with their own caveats [21, 111]: missingness, errors, and other flaws. And despite standard processes for dealing with caveats — recording the issues digitally, creating analog notes, or communicating directly with colleagues — these processes are separate and separated from tools for analysis.

We noticed similar behaviors during our own collaborations with domain scientists [95, 104]. Although we visualized the official datasets, the domain experts noted that the visualization did not reflect their mental models of the phenomenon under study. Interacting with the visualization caused the experts to externalize their knowledge about the many ways in which the data did not reflect the reality of their work. The experts drew on their deep experience and knowledge about the phenomenon to fill in the gaps of the data, which meant they were performing mental gymnastics every time they used a visualization for analysis. Other researchers have noted similar experiences — a missed opportunity to more tightly couple personal knowledge about data with visualization tools for analysis [115, 120]. Yet, we kept designing visualization tools that visualize first and foremost the data. This led us to reflect: *Why do we value imperfect and partial data, and devalue subjective expert knowledge? If we were to include expert knowledge in a visualization, how could we visualize it in relation to the data?*

Feminist epistemology drew our attention to the unquestioned assumption that visualization tools begin and end with data, despite recognizing the important knowledge experts had about the flaws and errors in their datasets. When we started to engage with feminist theory, we saw this tension and, in turn, shifted our attention to redefine the relationship between domain experts' knowledges and data. The main contribution of this chapter is a demonstration of how feminist theory impacts and alters visualization design. More specifically, with this work, we put forward a new concept *data hunches* that covalues expert knowledge and data as inputs for visualization design. This chapter's secondary contribution is a set of design guidelines and ethical considerations for incorporating data hunches in expert-led collaborative settings. In sum, these contributions illustrate how a novel concept driven by feminist theory offers a new approach to designing visualization tools by embracing data imperfection and partiality.

5.2 Data Hunches

Data hunches center personal knowledge as a critical component of data and visualization analysis. We define data hunches as *"an analyst's knowledge about how and why the data is an imperfect and partial representation of the phenomena of interest."* [95]. In other words, a data hunch is what the analyst knows about the data and the sources of its imperfections and entanglements.

[2] The interactive prototype can be found here: `https://vdl.sci.utah.edu/data-hunch/`

Data hunches are ubiquitous and critical for data-driven analysis and decision making. This phenomenon has been studied by other researchers [79, 110], but it is notable that often knowledge about datasets is captured *external* to the visualization tool in the form of note taking [101], annotations [152], and sketches [150]. This requires extra mental effort to use a visual analysis system. Thus, we advocate for data hunches to be visualized together with the original datasets to support a more holistic analysis session.

The conceptualization of data hunches builds on research about implicit error [104], uncertainty, and feminist epistemology. Uncertainty is an amply studied topic within visualization research. This is because it is readily accepted that there will be measurement errors, incomplete datasets, inaccurate model assumptions, and many other factors that make a dataset incomplete and imperfect. Yet, despite being a well-studied topic, there are a variety of definitions [21]. Some definitions emphasize the variability of possible values within the dataset [73], others focus on the dearth of information when creating the dataset [18], while alternative definitions focus on the limits of knowing something definitively [37]. We conceptualize data hunches in relation to uncertainty visualizations precisely because, as a topic, uncertainty centers the relationship between people, visualizations, and the limitations of data. Additionally, uncertainty visualization research brings forward the ways in which visualizations can help people work with the limitations of data through graphical interfaces.

However, the type of uncertainty addressed by data hunches is not easily quantifiable. As other researchers have noted: *"as the uncertainty becomes more abstract, it is more difficult to quantify, represent, and understand"* [141]. Likewise, our experiences working with domain experts yielded types of uncertainty resulting from unevenly distributed data production processes that resulted in a heterogeneity of errors in the data, impossible to model or fix systematically. This type of uncertainty is commonly referred to as qualitative uncertainty. Qualitative uncertainty is the subjective knowledge held about the ways in which data are uncertain. Researchers have identified the many and varied sources of qualitative uncertainty, like differences between interpretations of visual representations [21] or errors in the data that are not recorded but accounted for differently across individuals [104]. In contrast to visualizations of quantitative uncertainty — where the uncertainty can be modeled or represented numerically — the focus of visualizing qualitative uncertainty instead relies on eliciting, recording, and visualizing *sources* of uncertainty. For example, McCurdy et al. used a structured form to record expert knowledge about qualitative uncertainty in order to then visualize it as contextual information [104]. Franke et al. [58] also collected and visualized the experts' knowledge about their confidence of specific data sources in a visualization analysis tool. As these examples demonstrate, qualitative uncertainty — though framed as a lack of knowledge — is in fact known from the perspectives of experts who have deep knowledge about the data and its production.

Similarly, but stemming from a different epistemological perspective, critical data studies and feminist scholars have long critiqued that data are neutral and objective objects in the world. These scholars have refuted that data can ever be *raw* [22, 61] and have made the argument that *"taken out of context, data lose meaning and value"* [38]. Crucial to this perspective on data is that data are inseparable from and entangled with the social, political, cultural, and historical contexts in which they are produced. Entanglements shape data and give them meaning. Thus the knowledge that experts have about data are essential for sensemaking with data and the qualitative uncertainty that follows.

Data hunches takes the conceptual and epistemological step to regard data and knowledge as two different, but equally important, sources of *situated knowledge* [63] about some phenomenon in the world. Within feminist epistemology, *"the possibility of sustained, rational, objective inquiry"* [63] is reached through understanding the limits and partiality of any one *situated* knowledge and then by bringing these perspectives to achieve a more complete representation. By valuing expert knowledge as equally important to data in the visualization design process, data hunches can

lead to a more inclusive, complete, and rich representation of the phenomenon being visualized.

5.3 Operationalization of Data Hunches

In addition to building the underlying theory of data hunches, we also enumerated characteristics of data hunches that can be graphically represented as marks or channels. This operationalization was essential to clarify attributes of data hunches that could be easily translated and used within a visualization design. In other words, explicitly defining the *types of data hunches* that could occur and then enumerating how they could be recorded in *data, visual, and abstract spaces* makes the theory actionable. This type of translation work is necessary to employ ideas from feminist — and perhaps other — epistemologies in a computer science discipline.

To characterize data hunches, we iteratively discussed and reflected as a group on the different types of data hunches we had encountered with past collaborators. We used low-fidelity design sketches and a prototype implementation to prompt discussions and feedback on the kinds of data hunches we could record in a visual system and how the hunches related to the visualization and underlying data points. Our experiences regarding data analysis pipelines and visualization design drove us to synthesize these discussions into characterizations of data hunches that could support future design guidelines.

5.3.1 Characterizing Data Hunches

As part of characterizing data hunches, we identified and categorized the different types of data hunches we had encountered in our prior engagements with domain experts. This resulted in three categories of data hunches and three ways to externalize them within a digital system. These categories and precision levels are essential to operationalizing data hunches and incorporating them within visualization analysis tools.

The following are three categories of data hunches that speak to sources of trustworthiness, completeness, and accuracy.

Assessment Hunches: Assessment data hunches reflect knowledge about the trustworthiness of the data. This could include some sort of categorical rating about the data or a written comment. In Fig. 5.1, there are two ▭ that illustrate a graphical representation of assessment hunches.

Structural Hunches: Structural data hunches reflect knowledge about data items that should be included and excluded from the dataset. In Fig. 5.1, there are two ⬚ that reflect a data hunch about a missing category — Mexico in this instance — and a category (Canada) that should be excluded from the dataset.

Value Hunches: Value data hunches reflect knowledge about what are more accurate values for a specific data item. Value hunches are represented by ⬭ in Fig. 5.1 and are detailed further below.

We offer a more nuanced breakdown of value hunches. These three types of value hunches reflect the specificity of knowledge an individual may have and wish to externalize, which can be expressed numerically.

Directional Hunches: Directional data hunches support the externalization of knowledge that indicates a change in value but is not or cannot be specific. In Fig. 5.1, this is graphically represented by the arrow at the tip of Australia's bar chart, indicating that the value of Australia's COVID-19 cases should be higher, but without a specific indication of how much.

Specific Value Hunches: Specific value data hunches support the externalization of knowledge that is a specific and different value from what is recorded in the data. In Fig. 5.1, the arrows

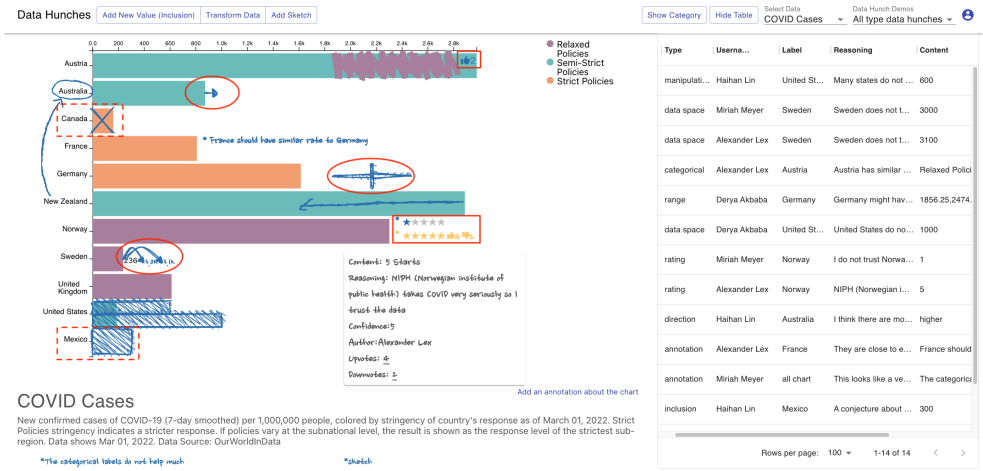


Figure 5.1 — Screenshot from data hunches prototype that illustrates the graphical implementation of different characteristics of data hunches with a dataset of confirmed COVID-19 cases across several countries as of 1 March 20222. The ▭ represents different ways to externalize *assessment hunches* through star rating systems and thumbs up/down voting affordances. The ⬚ represents both inclusion and exclusion structural hunches. In this example, Mexico is included in the visualization, while Canada is removed. The ◯ represents *value hunches* indicating that the actual value should change in direction, specific value, or fall within a certain range.

pointing out of Sweden are associated with particular numbers, indicating that the data hunch is more specific.

Value Range Hunches: Value range data hunches support the externalization of knowledge that is somewhere in between directional hunches and specific value hunches — the hunch is captured in a range of values. In Fig. 5.1, there is a range next to Germany's bar — this range indicates some uncertainty in the data hunch but still offers a bounded range of values that reflect the data hunch.

5.3.2 Recording Data Hunches

This section outlines the data, visual, and abstract spaces to further characterize data hunches. Figure 5.2 provides a visual overview of the techniques we suggest for recording data hunches. The techniques are grouped into data, visual, and abstract spaces as they pertain to visualization analysis and how one may record and visualize data hunches.

Data Space One way to record a data hunch is by suggesting an alternative data point or model. Using *form-based manipulation*, an analyst can enter their data hunch as a value or other data point directly through a table or form. Using *model-based manipulation*, the user can create a model to record their data hunch about the general behavior of the data if it follows some mathematical function. Recording data hunches in the data space supports storing the hunches in a data format, which could make it easier to analyze.

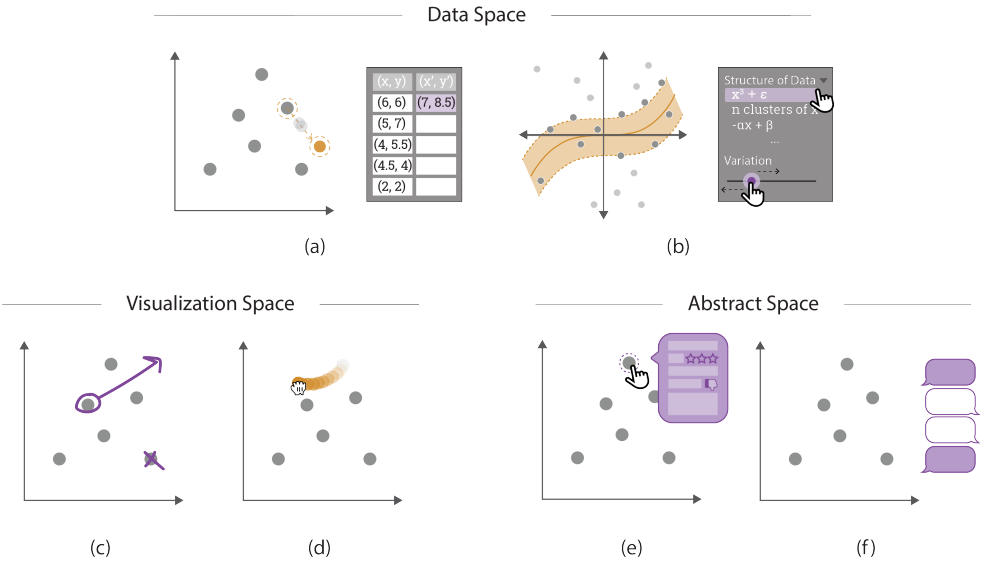


Figure 5.2 — A visual overview of abstracted techniques for recording data hunches in a visual analysis tool. There are three spaces in which these techniques fall: the data, visual, and abstract space. In the data space, data hunches can be recorded with (a) form-based manipulation or (b) model-based manipulation. These techniques are appropriate when a value or a model can represent the data hunch. In the **visual space**, data hunches can be recorded with (c) free-form sketching or (d) direct manipulation. These methods allow analysts to record their data hunches visually, useful when they may not have a specific value in mind. In the **abstract space**, data hunches can be recorded with (e) structured elicitation or (f) textual annotations. These techniques allow analysts to record their data hunches that cannot be captured through techniques in the data or visual space.

Visual Space The next way to record data hunches takes advantage of the unique visual and interactive properties of a visualization. In this space, experts can record their data hunch using *free-form sketching* to add drawings to the visualization. This method allows analysts to visually mark up the existing visualization, similar to how one would with a whiteboard and marker. With this technique, however, the data hunch cannot be converted into data easily. Another method is *direct manipulation*. This technique supports recording data hunches through interaction with visualization elements, like resizing a bar in a bar chart or dragging a point in a scatterplot. With this technique, the data hunch can be recorded as structured data, but is limited to the specific modality afforded by the chart type.

Abstract Space The last space we put forward is abstract because the data hunch cannot be expressed through data or visual spaces. Two techniques are *structured elicitation* and *textual annotation*. Both techniques rely on recording data hunches through text inputs that are either predefined or open-ended. This space also supports reactions to other hunches, like voting or commenting.

5.3.3 Designing for Data Hunches

In addition to characterizing data hunches, we present design guidelines for incorporating data hunches in visualization systems. The guidelines are derived from our iterative design processes, both with the low-fidelity sketching and the interactive prototype. While the characteristics of data hunches present a map of possible data hunch types, these design guidelines support decision making in how and where to graphically represent data hunches. The guidelines also embody the values of both foundational visualization theories and feminist epistemology, striking a balance between designing an effective visualization, ethically recording hunches, and intuitively differentiating data from data hunches.

DG1: Do not change the original data. Data hunches and the original dataset should remain separated in order to uphold the legitimacy and quality of the underlying data. This guideline should be enforced both graphically and technically — taking care to keep data hunch records in separate datasets.

DG2: Make data hunches distinct. Falling from the prior guideline, using different encodings between data and data hunches ensures a clear delineation between the two. In the prototype, we used a sketchy rendering library [157] and handwritten fonts to distinguish data hunches from the clean, unadorned lines of the original data visualizations. This is not the only visual metaphor that can be employed. However, we caution against encodings that can either obscure the underlying data or be confused as an attribute of the original dataset.

DG3: But, make data hunches similar. It remains important to maintain some graphical similarity between data hunches and the underlying data to more closely couple analysis using both sources of information. With this guideline, we suggest using similar visual channels to support comparison across data elements and data hunch elements.

DG4: Keep data hunches close. In line with choosing similar visual encodings, data hunches should be visualized near the visual space they describe. For instance, if a hunch describes the legend, then a graphical representation of the hunch should be placed next to the legend. Keeping hunches close to the parts of the data they describe ensures that the information from data and data hunches is used together during analysis.

DG5: Use direct manipulation. This design guideline extends the recommendations of keeping things similar to interaction design choices. Using interaction cleverly can reinforce the notion that data hunches originate from interactions with data. Using direct manipulation can give the feeling then that the analyst is directly moving or annotating the underlying visualization, while also providing immediate visual feedback.

DG6: Design for data hunches. The decision to incorporate data hunches will alter how a visualization is designed. Considerations for appropriate marks and channels for the original dataset are not enough to adequately account for the addition of data hunches. One such consideration is how to support the addition of specific value hunches. With certain visualization types, like pie charts, adding one hunch will inevitably affect other data points, given that such visualization techniques use area segmentation. Another consideration is maintaining empty space for annotation and manipulation. Incorporating extra white space supports assessment and structural hunches, similar to how margins in textbooks hold marginalia. Ultimately, considerations for layout and additional space are necessary when designing for data hunches.

DG7: Design for collaboration. As data hunches play a role in collaborative analytical roles, it is important to design for collaboration across multiple users. Some design considerations are user-specific logins and interaction with other hunches in the form of endorsements or commenting.

DG8: Elicit context and accountability. Informed by feminist epistemology, at the core of data hunches is the recognition that context is critical to knowledge production. As such, we advocate for recording the relevant context of data hunches. This could take the form of

recording *who* created the hunch. Or, it could be a prompt regarding *why* they had the hunch. Further, we imagine including a host of other context-eliciting metrics, like confidence level, the author's seniority level, or the individual's experience working with the dataset.

5.4 Considerations for Using Data Hunches

Working with feminist epistemology as our theoretical frame kept questions regarding ethics and data at the forefront of our discussions and prototype development. We see many potential opportunities for using data hunches and also the challenges that come with designing explicitly for technologically-mediated collaborative knowledge production.

Maintaining Responsibility This work was conducted during the COVID-19 pandemic and amidst a rise in using visualizations for disinformation [92]. We deliberately scoped the discussion of data hunches to a group of expert collaborators with extant trust and little incentive to misuse data hunches toward personal gain. However, technological use is hard to control once released into the world. Therefore, it is important to note that a risk with data hunches involves using technology to modify a dataset for persuasive reasons. That said, research also indicates that social discussion in online communities positively contributes to social sense-making [62, 80]. So, how do we maintain responsibility between misuse and collaborative sensemaking? A feminist ethics maintains that we *"stay with the trouble"* [63] — rather than limiting data hunches to a select group of individuals, we argue that data hunches can and should be tested in public settings, but with a recognition that it may require additional design considerations to reduce potential harm.

Tackling Identity Another tension revolving around the use of data hunches is the complexity of identity. Identity — in the form of a user name or persona — is one of the many contexts encoded within data hunches and can contribute to how much another person trusts new information [82]. Revealing identity though leads to a host of questions on how to protect vulnerable people with unpopular or contradictory hunches. How could we then design to support data hunches but also maintain aspects of identity that build trust in the hunch?

Addressing Trust Closely paired to the issue of trusting another individual's data hunch is the effect of trusting the underlying data in the presence of many conflicting data hunches. It is unclear how an individual would respond to a visualization filled with data hunches that call into question the integrity of the original data. Recent work has tested this idea in experimental studies [80]. It reveals a general interest of individuals to engage with conversations about the visualization and data, but it remains to be seen what would happen in an unmoderated setting. Perhaps, data hunches will become akin to uncertainty visualizations, a feature that will be excluded to reduce complexity [73]. At the heart, though, data hunches formalizes an alternative approach to developing collaborative visualization analysis tools that requires more testing.

Recording Data Hunches as Data The final consideration we put forward is the idea that a system designer could record and then use the data hunches as another dataset. Although this idea is implicitly embedded in the design guidelines — *do not change the original data* and *make data hunches distinct* — it remains unexplored how a dataset generated from data hunches would alter and support visualization analysis. If data hunches were tracked over time, it could be possible to visualize how individuals' knowledge about the data changes over time. Or data hunches could perhaps augment analysis across different versions of the dataset, acting as scaffolding for interactive visualization analysis. However, is this something we *should* do? Are data hunches epistemologically meaningless without the context and data that created them?

5.5 Reflections for the Dissertation

This chapter demonstrates how feminist epistemology changed design constraints and resulted in novel design guidelines. Specifically, we put forward situated knowledges as an alternative perspective on data, which shifted our focus to the entanglements of data. These entanglements represent not only the mechanisms and processes that contribute to the creation of a dataset, but also the relationship that experts have with their data and their deep knowledge of these entanglements. This new perspective brings forward expert knowledge as an essential input to visualizations akin to data, challenging what belongs in analysis tools. A sensitivity to situated knowledges and the entanglements of data requires accountability and responsibility to design for all sources of information, not just data.

This chapter contributes the concept of data hunches, which encapsulates the entangled relationship between data and the domain experts who use that data. We concretize our conceptualization by specifying the data, visual, and interaction spaces of data hunches within visualizations. Further, we put forward design guidelines that draw on our experiential knowledge and informative discussions.

Concerning this dissertation, this chapter represents how feminist epistemology changed how we approach and design visualization systems. The work on data hunches picks up on prior work by McCurdy et al. [104] who had difficulty reconciling the errors present in official data sources with the domain experts' rich, contextual, yet subjective knowledge. Rather than design a system that lifted the knowledge of the experts, the resulting visualization tool design kept that information as secondary to the data. In hindsight, we recognized that the project was limited by the epistemology of the work, unable to reconcile *objective* data with *subjective* expert knowledge. Our later work theorizes about this moment, calling it an epistemic wall [5].

We argue that visualization researchers should recognize epistemic walls because visualizations reflect and instantiate the perspectives taken by the researcher and designer, thus perpetuating a specific world-view [47, 68, 70, 85]. The work presented in this chapter reexamined the assumptions of designing visualizations for experts guided by feminist theory, highlighting knowledge's situatedness. Reframing data as equal to, rather than the prime source of truth, supported design decisions that brought forward expert knowledge as first-class data objects to be visualized *with* other data sources. With this work, we demonstrate the capaciousness for alternative epistemologies to progress visualization design while co-existing with dominant epistemological stances.

This chapter also demonstrates the necessary *bridging* work required to bring a new concept like data hunches to visualization design processes. Beyond the theory and definition of data hunches, relating data hunches to data attributes made it possible to incorporate data hunches in a visualization design process. Further, the guidelines we derived straddle two epistemological spaces of rigor: one of positivist origins and the other of feminist. Positivist rigor relies strongly on reliability and reproducibility through data analyzed with the intent of *"producing an unambiguous result that is representative of reality"* [107]. To that end, data should remain preserved in its original form and all further modifications recorded faithfully. Feminist epistemology does not disregard data but focuses instead on the constructed and relational nature of reality. Thus, knowledge is subjective, dependent upon the context from which it is created. And so our theorizing of data hunches brings these two epistemological stances to a head within a visualization design.

CHAPTER 6

Ethics in Collaboration

In this chapter, we review the impact of applying feminist epistemology to visualization frameworks of collaboration. The core of this chapter focuses on an interview study with constellations of visualization researchers and their domain expert collaborators. The purpose of the study was to understand the lasting impact of visualization collaborations. By drawing on a feminist theory, our attention shifted to examine collaborations through a set of relational, emergent, and context-specific ethics. This reexamination revealed that extant visualization frameworks lack recommendations for considering maintenance of tools and power dynamics within collaborations. This chapter is based on the following published work:

> Derya Akbaba, Devin Lange, Michael Correll, Alexander Lex, and Miriah Meyer. *Troubling Collaboration: Matters of Care for Visualization Design Study.* ACM SIGCHI Conference on Human Factors in Computing Systems, 2023.
>
> Supplemental Material:
> https://osf.io/cvrbs/

> Derya Akbaba and Miriah Meyer. *'Two Heads are Better than One': Pair-Interviews for Visualization.* Proceedings of IEEE VIS (Short Papers), 2023.
>
> Supplemental Material:
> https://gotdairyya.github.io/pair-interview-primer/

Backstory

This chapter mainly draws on the work presented in *Troubling Collaboration.* Everyone on the team had experience working with domain experts in standard visualization collaborations, like design study [137]. The team had a range of experiences reflecting our roles as current and former students — responsible for the daily interactions with collaborators — and project leads — responsible for initiating the research, supporting the student, and ensuring deliverables are met. Despite the varying experiences and responsibilities, we all felt uncomfortable about how collaborations ended. We were plagued with questions like: *What, if any, lasting impact*

did our collaboration have on the people we worked with and their research? And what was our responsibility to our collaborators and the tools we built for them to follow up and offer maintenance? We addressed our concerns about impact and ethical responsibility through this work. I turned to feminist theories and implementations of care ethics and maintenance to scaffold discussions of how visualization researchers could collaborate differently. Meanwhile, Devin Lange and I carried out a series of interviews, learning and refining our interview method along the way. We published a short paper on the interview method in *'Two Heads are Better than One'* in order to formalize the benefits of conducting interviews with two people.

6.1 Case Study: Troubling Collaboration

Visualization research is often conducted in applied and collaborative contexts. Design study (DSM) [137] is a common methodology and method of inquiry within visualization research. The methodology is so ubiquitous that in 2022, the paper received the Test of Time Award at the IEEE Visualization Conference and has been cited over 1157 times [1]. While DSM provides a comprehensive framework for how to establish and collaborate with domain experts, it lacks guidance on how to engage with collaborators *after* the project is complete. This gap opens up many questions regarding the *ethics of exit* [107], the responsibility of maintenance, and the long-term impact of visualization collaborations. This gap is exacerbated by a field and funding agencies that focus on and reward novelty, as opposed to ongoing projects or failures, resulting in little archival material and longitudinal studies that report on impact over time.

To understand the impact of visualization collaborations, we conducted an interview study with 20 visualization researchers and their collaborators from differing perspectives: visualization graduate students (6), visualization faculty (8), and domain experts (6). We asked them both about their experiences during the collaboration, along with what happened afterward. We applied a feminist epistemological lens — *matters of care* [40] — to understand the entanglements of agency, power, and domain expertise within collaborations. Engaging with feminist theory shifted our attention to dynamics between individuals in the collaboration, neglected people and technology, and how care is unequally distributed along axes of power. The result was a complex and, at times, contradictory web of motivations and actions. This entangled web troubled our initial thoughts about what it meant to care and consider impact within visualization collaboration.

The main contribution in this chapter is a demonstration of how feminist epistemology attunes our sensitivities toward ethics in technology design and research processes, and consequently troubles how care is presented and valued within collaborative frameworks. In support of this claim, we present an interview study analyzed through a lens of care that highlights concerns of long-term maintenance and power asymmetries embedded and inherent to how design studies are conducted. We contribute two new methods in this study as well: *pair-interviews* and *diffractive reading*. Pair-interviews expand the set of sparse interview methods within visualization. Diffractive reading serves as another example of how to operationalize feminist theory in qualitative analysis. These secondary contributions are furthered by a set of provocations derived from feminist theory that aim to establish a different way to collaborate, supporting the idea that design study can be run care-fully.

[1] Number pulled from Google Scholar on 12 May 2025.

6.2 Case Study Methods

For this work, we sought visualization researchers who had conducted collaborative research projects with other researchers. We sourced an initial participant list from papers that had cited design study in their methods section, which resulted in a list of 156 visualization researchers and 79 domain experts. We sent out an initial survey as a recruitment tool: it contained 17 multiple-choice questions. Those who completed the survey could receive a 10 USD gift card or could choose to donate it (92% donated). Of the total contacted participants, only five agreed to a follow-up interview. We sourced additional interview participants through convenience sampling. The sampling resulted in 17 additional interview participants who had conducted collaborative visualization research.

6.2.1 Interviews

We conducted *pair-interviews* [2] over the course of five months. The interviews were conducted in English over a videoconferencing platform. They lasted about one hour, and participants could choose between 20 USD gift cards or again, to donate the same amount (95% donated).

The pair-interview method — as outlined in our paper [2] and a secondary contribution of this dissertation — is an interview method that has two interviewers supporting one another. This is a novel method because it explicitly introduces the use of a two-person interview team for a visualization context. Each researcher has a distinct role during the interview. The first role, the driver, is responsible for moving the interview along by asking questions and engaging with the interviewee. The second role, the navigator, keeps the interview on track by managing time spent on any one topic and asking follow-up questions that the driver may have missed.

Our interviews followed an interview script that covered four topics of interest: the logistics of the collaboration, how the collaboration ended, how decisions were made, and how the interviewee felt about the collaboration. These questions reflected our research goals of better understanding care within these collaborative projects. After the interviews, the audio files were transcribed and anonymized through a mix of automated and manual methods.

The corpus of interview transcripts comes from 20 interviews conducted between January 2022 – May 2022. The transcripts reflect interviews across different roles and responsibilities in visualization collaborations. The interviews represent three different constellations of perspectives. There were 4 *triads*, which meant we heard about the collaboration from the perspectives of a visualization PI, their graduate student, and a domain expert. Next, we had 6 *dyads*, representing just the students' and domain experts' experiences within a collaboration. And finally, we interviewed 4 independent PIs who tended to discuss multiple collaborative projects during an interview. Our interview participants came from North America and Europe and ranged in seniority across roles. The domain expert collaborators worked in academia, companies, and non-profits, representing STEM and humanities fields.

6.2.2 Diffractive Reading

The interviews were analyzed using *diffractive reading* [7, 8]. As a nascent method with only a few examples in HCI [87, 91, 133], the underlying theory from Barad is operationalized differently. As a secondary contribution, we offer our interpretation of diffractive reading as an attention to differences across the interviewees' experiences and the differences within the research team analyzing the interviews.

The interviews were analyzed first individually, then as a categorical set: triads, dyads, or principal investigators (PIs). There were eight discussion sessions that lasted 2–3 hours each. Our varying

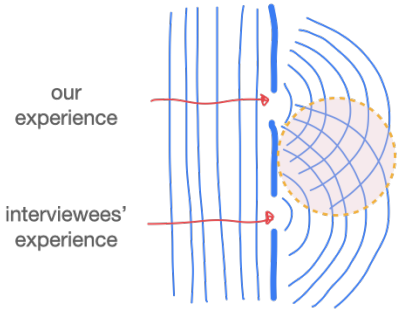


Figure 6.1 — Symbolic representation of diffractive reading. Drawing on imagery of diffraction, the interviews were interpreted at the overlap of our own experiences and those of the participants. At this overlap, we paid attention to the *differences that mattered* and why they mattered.

perspectives — one graduate student who conducted design studies and two senior researchers who have engaged with design studies personally and through students — led to a rich discussion and interpretation of the interview transcripts. While our personal experiences drew us to interact with the participants' stories in different ways, matters of care centered our discussions around stories of neglect, maintenance, and distributions of power.

6.3 Case Study Results

We focused our analysis on two main topics that surfaced across the interviews: neglect and power. Matters of care attune us to how care for one thing results in neglect of another [40]. Through a lens of care, we saw how the maintenance of tools is a neglected thing within visualization process models [103, 105, 137]. Stories from the interviews revealed more nuance — while missing in the process models, maintenance occurs in practice as acts of care. The second salient theme across the interviews concerned power asymmetries. Interpreting the interviews through a lens of care brought forward the ways in which visualization collaborations are embedded with relationships that are equally shaped and altered by project goals and power dynamics. The interviews revealed entangled motivations, objects of care, and feelings of agency. Here we present only a brief summary of the results.

6.3.1 Neglect and Maintenance

We chose to highlight three major themes regarding maintenance: the responsibility of students to maintain, the ways in which access to resources affected maintenance, and the benefits of collaboration beyond a tool. Concerns about maintenance were intertwined with personal responsibility, seniority, and access to resources.

Graduate students conduct most collaborations, and therefore, most tools are also developed by students. When asked about the responsibility of maintenance after the collaboration ended, two students spoke about the personal relationship with the collaborators driving their motivation

to maintain the tool. Though domain experts recognized the labor of the students, their work often remained invisible to their supervisors. Other students put their code on GitHub, citing the open-source code as maintainable by anyone, not only themselves. Across graduate students, there was no clear answer on who should maintain and for how long. All students expressed care, but dealt with limited resources and the burden of responsibility in different ways.

In some interviews, we were exposed to the different infrastructures built to support maintenance. In two projects, there was access to a third-party software engineering team, greatly reducing the responsibility of the individual student to develop and maintain a visualization tool. This, though, was the minority. In other interviews, PIs bemoaned the fragility of technology and shifting research goals that make it difficult to prioritize maintenance. Transparently, PIs discussed the many socio-technical factors that make most tools not worth maintaining: students graduating, changing tech stacks, or collaborators no longer using the tool.

Despite this inauspicious view on visualization collaboration outputs, many domain experts discussed the ways in which they benefited from the collaboration beyond the tool. Some spoke of learning new things about their data because explaining it to the graduate student often led to new points of reflection. In other instances, the interdisciplinary nature of the collaboration brought new perspectives to the domain expert's work. And during other interactions, the collaborations were a medium for collaborators to learn more about visualization research and the unexpected potential to use it for their own research.

6.3.2 Power and Objects of Care

A generalized premise of visualization collaborations is that they are transactional: the collaborator provides time, data, and interesting domain-based research questions, and in exchange, visualization researchers provide a custom tool. This results in a long period of unbalanced reciprocity, where collaborators are giving, perhaps less than, what they are receiving from visualization researchers. On top of this, most collaborations are conducted by graduate students who are funded and advised by visualization PIs who have the power to sway the direction of the research. Matters of care highlighted the many ways in which the structure of design study is embedded with power asymmetries. Like maintenance, the interviews provide complex stories of power, limited attention, and the fallout of some things receiving care while others are neglected.

One area of tension occurred between the desired output of visualization researchers — a novel tool with an accompanying publishable paper — and the collaborator wanting a useful tool, where useful became contentious when it did not beget novelty. Despite this, there were visualization teams that put the needs of the collaborators ahead of a novel tool, expressing care and regard for their collaborators' time. In other constellations, this was not the case, demonstrating the power imbalance in the collaboration. While the visualization team received feedback on prototypes, collaborators sometimes saw this work as an effort misused among their regular workloads. There were similar expressions of frustration when it came to sharing datasets with the visualization team or interpreting data for a paper: it was a lot of work that aided the visualization researchers but not the collaborators. In both instances, collaborators spoke of misaligned goals and tasks requested by the visualization team.

In our interviews, we also explored power asymmetry within visualization teams. PIs reflected upon their power and responsibility to mentor successful PhDs while also ensuring project goals were met. At times, these goals were not aligned. Collaborations often lasted longer than single PhDs and involved deliverables to the funding agency. Some PIs discussed how balancing these promises with the student's research interests put them in difficult positions — acting as a mediator between previously agreed upon outputs and the student's decision to lead a project in an unexpected direction. Essentially, PIs were responsible for the success of their student, the success of the collaboration, and their own longer-term research goals. While care has a place

across all these responsibilities, there is little guidance regarding where, when, or how it applies to design study.

6.4 Implications for Visualization: Care-ful Design Study

With this work, we bring forward the ethical considerations that arise from acknowledging entanglements in visualization collaborations. Entanglement theory specifically and feminist theory more broadly draw attention to the dynamic web of politics, power, and relationships that shape and influence the outcome of collaborative visualization research. Caring about these dynamics and relationships foregrounds an ethical approach to knowledge production embedded in and inseparable from our research process. Care directs attention toward certain things and not others, resulting in a responsibility for researchers to understand how their care is also entangled with neglect. By embracing entanglements, we present design study as a locus of varying interests, agency, and autonomy. The following provocations bring forward matters of care for design study, with the goal of moving toward care-ful design studies.

6.4.1 Results of the Study: Care within Visualization Teams

We present two provocations for visualization PIs to consider before engaging their students in a design study. The provocations are directed at PIs because they secure funding, establish relationships with collaborators, and advise students. In other words, they have the agency and power to decide when to conduct a design study and when not to.

[P1] *Will the student benefit from the design study?*

[P2] *Can you support a care-ful design study?*

Current guidelines for design study point to many necessary skills that researchers must employ: designing, engineering, and employing qualitative methods and more [107]. However, our interviews also demonstrated that there are many other skills that support design study: understanding people and organizations, navigating and negotiating across competing interests, and establishing trust and rapport with individuals. All of this combined can be difficult to manage, especially given that graduate students often have very little agency. And so the question remains: is design study appropriate for graduate students? Despite the many interpersonal and structural challenges of conducting a design study, there are also many benefits that students spoke of, like enjoying the project, making an impact, and learning new skills. Given both challenges and benefits, we suggest that PIs pay attention to power asymmetries within the collaboration to find opportunities to increase the students' agency or offer care and support.

Conducting care-ful design study asks PIs to reflect on how far their care can extend. In other words, what are the ways they will engage with social, technical, and organizational entanglements — supporting students and collaborators with their own goals in addition to their research and funding objectives. But care is also a reminder to consider what is also neglected. As we saw across interviews and process models, there is a lack of guidance for how to maintain tools after collaborations end. Who will maintain the tool and at what cost?

When responsibility for maintenance is not specified, it affects graduate students and collaborators. Maintenance is complex, extending beyond the technical expertise expected from students: it requires time, energy, and motivation. Through a lens of care, we can envision new futures for

visualization tools. One alternative future is to consider how to make components of visualization tools reusable by other researchers. Rather than a junkyard of broken tools, we move toward a reuse center of possibilities. *Designing for the graveyard* is another such future that reframes visualization prototypes not as fully engineered products but rather as methods of inquiry and mediums for feedback [70, 107]. This opens up the space for more experimental designs that play with complexity [84, 99], sketches [150, 151], and physicalizations [118, 148]. By embracing this perspective, a care-ful design study establishes visualization prototypes as research tools rather than fully functioning and engineered tools.

Another consideration of a care-ful design study is reconsidering whether a novel bespoke tool is necessary for the collaborator. Although novelty is an important criterion for publishing, tools require organizational and social maintenance, beyond the scope of any one student or collaboration. So, how could we meet our collaborators halfway? We suggest developing plugins or visualizations that fit within the collaborator's current technological ecosystem. Designing with these constraints could result in visualization tools that are more sustainable and used more widely.

6.4.2 Results of the Study: Care for Collaborators

To conduct design study care-fully requires reflection about the types of care we hold for our collaborators. With that, we bring forward two provocations for the visualization team and broader visualization community.

[P3] *Are we heroes or are we collaborators?*

[P4] *Are our collaborators truly collaborators or participants?*

A recent critique of visualization research has characterized the work as heroic [34], offering an exceptional visualization tool as compensation for domain experts' time, data, and energy. This perspective, reflected in the design study framework [137], focuses on the output of visualization collaborations. In contrast, a care-ful design study emphasizes that visualization researchers can contribute incrementally by paying attention to the needs of their collaborators beyond a bespoke tool. Some opportunities to enact this include sharing visualization tips and tricks, supporting collaborators with their data management and analysis, or producing more standard static charts for publications or presentations. This shift moves away from a heroic tool toward enacting care in the moment.

With care-ful design study, we also see the importance of communicating goals and expectations across a design study. Over the course of a design study, domain experts were sometimes collaborators — shaping research goals with the visualization team — and at other points were treated as participants — providing feedback and testing prototypes. The latter role left domain experts powerless and, as a result, frustrated. Through open communication, supporting collaborators' goals incrementally along the way, and acknowledging the domain expert's labor, we can reduce the harm inflicted by power asymmetries within the collaboration.

6.5 Reflections for Dissertation

The work presented in this chapter demonstrates how approaching a familiar visualization method of inquiry — design study — from a different epistemological perspective revealed important gaps in the existing frameworks. Through the lens of feminist care ethics, our attention shifted to pay attention to the relationships present within the collaboration and with other influences like power, access to resources, and agency. Our change in perspective revealed that extant

frameworks lacked guidance on what to do *after* a collaboration ended and how to care for collaborators *during* the collaboration. Said differently, we found that maintenance is neglected, or rather, omitted from these frameworks.

Further, the work presented in this chapter demonstrates that ethics is inseparable from research processes and technology development. Entanglements directs our focus on the relationship between people working together in different capacities, as well as the decisions that impact what is studied and what is not. From this perspective, decisions regarding who to work with and what topic to work on also implicate who and what are neglected. As a result, ethics becomes entangled with decisions to develop visualizations and pursue collaborations, which is a shift away from *post-hoc* considerations of ethics. Entangled ethics moves the onus from individuals to collective negotiation, communication, and accountability. Emergent, relational, and situated ethics beget researcher reflexivity that considers both historical and future-looking responsibility, driving our research to consider visualization research beyond tools.

CHAPTER 7

Discussion & Future Work

> *Rather, the point is that these* entangled practices *are productive, and who and what are excluded through these entangled practices matter.*
>
> –Karen Barad [7, p58]

Following Barad, the discussion focuses on who and what are excluded in this dissertation. These entangled exclusions are presented as future work — areas where we see entanglement theory for visualization and alternative epistemologies broadly impacting visualization research.

7.1 Bridging Disciplines

To develop the epistemological theory presented in Chapter 3, we worked closely with Lauren Klein — my co-advisor and humanities researcher — for many years. This collaboration provided the groundwork to learn about entanglement theory's historical and intellectual lineage. It was a productive exchange of ideas across disciplines, resulting in a published paper at a visualization venue [5]. A lingering question remains, though: *Was the benefit of collaboration mutual?* Other fields have stronger and longer histories of grappling with epistemology, leaving us to wonder if there are any possibilities for us, as visualization researchers, to contribute to other disciplines beyond developing a visualization tool?

We partially explored the answer to this question in Chapter 6, where we report on the results of an interview study with visualization researchers and their collaborators. In this study, some collaborators remarked on how the collaboration changed their approach to their research. One collaborator spoke about how the collaboration with visualization researchers provided a different ontological and epistemological perspective on their data, providing a new data-oriented path of inquiry. Another domain expert discussed the unexpected benefits of new research methods and tools for communicating their research that came from the collaboration. In these examples, visualization's unique disciplinary values, methods, and epistemologies shaped and altered another discipline beyond the confines of an analytical tool. What would it mean for visualization researchers to deliberately conduct this disciplinary bridging work? Visualization research holds

a unique perspective on the workings of data, visualization, interaction, color, context, and more — what would the consequence be of articulating these ideas beyond implementing technology?

One path forward is to articulate explicit epistemologies for visualization research. Visualization lies at the core of many epistemological theories. In their historical account of objectivity, Dalston & Galison discuss how scientific images have changed over time to substantiate claims of objectivity [39]. Feminist scholar Donna Haraway draws attention to the power of visualization practices that concretize specific versions of truth over others [63]. If we were to follow such a path, it would be with the acknowledgment that it blurs the lines of philosophy of science and visualization research. The potential, though, is that it could make the knowledge-making practices of visualization explicit so that it is legible externally. Correll marks a similar path forward, advocating for a *"strong and shared epistemic and theoretical foundation for visualization work"* [35]. The only difference between our suggestions is that we advocate for a plurality of epistemologies, which we expand on more in Sect. 7.3.

Another path forward is to address gaps when we see them. Visualizations are used ubiquitously, but it is less clear where visualization research has left a mark. Again, drawing on the interview results from Chapter 6, there are other ways to contribute beyond a visualization tool. We could actively seek and support our collaborators regarding creative methods, working with data, or making graphical representations for reports. These visualization applications may seem simple, but as long as there are gaps, they signal research opportunities to bridge them. In the next section, we continue this thought, expanding on the role of visualizations *beyond* academia.

7.2 Entanglement Visualization for Social Impact

In one of the first papers to draw attention to the theoretical foundations of visualization, van Wijk asks: *"are we, as researchers in visualization, on the right track?"* [149]. He goes on to articulate that *right track* means articulating and assessing the impact of visualization research beyond academia. This concern is repeated years later by Correll, who reports on senior researchers expressing that there is a *"lack of concern for potential real-world uses of visualization research"* [35].

To shift from academic research to real-world impact, we put forward entanglements for visualization as one suggested path. Entanglements is grounded in the ways that power, privilege, and society affect knowledge contributions, emphasizing relationships as sites of knowledge production. Leaning on entanglements for visualization, research about visualizations for social impact would look to the dynamic and constitutive relationships between people, data, and other influences to understand how visualizations work in the world. Although this type of inquiry is already happening through methodologies like design study [137], the application areas primarily focus on scientific or organizational disciplines with large quantities of data. What would it mean to study visualizations when the datasets are small? Or, when the users are less interested in precise analytical insights? How do the entanglements of people, their background, and their relationships to data and visual representation change how they use and want to use a visualization? We speculate that entanglements for visualization is well-suited to draw our attention to the unexpected influences of visualization use in heterogeneous, social contexts.

Toward these questions, we have begun to explore the entanglements of conducting visualization research in social-oriented contexts. We've been collaborating with a local nonprofit — Östergötlands Stadsmission — that supports marginalized and vulnerable communities across Sweden through various initiatives. We've been working together to co-develop workshops that use data as a tool for reflection, communication, and empowerment.

In the spring of 2024, we developed and ran a series of three pilot workshops with one of Stadsmission's youth leadership programs. The workshops followed a similar format: they were

all data physicalization workshops centered on one question chosen by our collaborator. As part of the workshop, the participants created datasets about the central question and made data physicalizations. The workshops ended with group discussions about the topic.

A wealth of unexpected entanglements has fallen from this collaboration. Some of these entanglements stem from the contexts of running the workshop, and the others involve the level of engagement with data collection and visualization. Though it is beyond the scope of this thesis, conducting visualization research in a social context has introduced a host of challenges that have received little attention in the visualization literature. The entanglements of this work has left us to grapple with the ethics of creating sustainable, long-lasting material that will exist beyond the research collaboration, visualization activities that can engage across varying levels of data and visual literacy, and contemplation about the appropriate time and place to use digital visualization tools.

7.3 Translating Theory to Visualization Practice

As the final point of discussion, we revisit the topic of translating theory to visualization practice. In Chapter 5, we discuss the bridging work necessary to move from the theory of data hunches to a concept that can be operationalized for a visualization system. This translation process required iterative design and prototyping work that drew on our wealth of experience working with data and developing visualization tools. Despite the generalized characteristics and design guidelines we provided, there is undoubtedly more work to be done in developing systems that can support data hunches in real-world data analysis scenarios. The design of a simple bar chart quickly became complicated with the addition of data hunches. Additionally, we had to change our approach to designing the visualization to accommodate for annotations and additional graphical elements. The visualized dataset was also relatively small and easy to interact with within one view. This implies and emphasizes that future theoretical and technical work is necessary to scale data hunches to visualization tools with multiple collaborators working on complicated datasets with complex linked-view visualizations.

The limitations of scaling data hunches reflect a broader concern regarding how to apply theoretical methods in practice. We are not the only ones who have pointed out the complications of translating theory to visualization design. In the work presented by Kim et al., the authors chose a simple visualization as a testbed for Bayesian elicitation [82]. The authors remark: *"even this simple scenario required reflecting on the best way to frame the elicitation of a prior."* They offer suggestions on the work it would take to apply Bayesian modeling to more complex visualizations. In another line of research, Yang et al. reflect on the complicated set of decisions required to develop election polling visualizations that communicated uncertainty *and* functioned outside a laboratory setting [162, 163]. The authors comment that their *"primary challenge was to connect a theoretical design space with the practice of making real visualizations that people actually understand"* [163], illuminating the complexity of design decisions and research that were necessary to achieve this goal. These examples demonstrate that the steps to bring theoretical conceptualization into implementation are non-trivial.

Another space that could see potential benefits from engaging with alternative epistemologies is scientific visualization (SciVis). The gap here, however, extends beyond epistemology, but rather incorporates receptivity. As one of my colleagues kindly said to me: *The visualization you talk about is not the same as mine.* Her comment highlights a vast space between applying feminist epistemology in my research and something actionable in her daily work. This is a similar response we have heard to the 2024 VIS capstone by Catherine D'Ignazio, one of the co-authors of *Data Feminism* [42] — while the ideas of data visualization conducted from a feminist perspective is interesting, it is challenging to apply it to visualizations that are not

explicitly about society and people. Stated differently, there is a translation gap between feminist theories and their application in visualizations about spatial, quantitative, and *scientific* data. This leads to a host of exciting opportunities for future research. What would a feminist scientific visualization look like? What are the unique entanglements that come from working with spatial data? How would our perspective shift if scientific visualization employed methods informed by feminist epistemologies?

There is clearly more work to be done to translate new theory across different visualization contexts that is actionable. With this discussion, we aim to draw attention to the *in-between* work that brings theoretical methods to practice as essential and indispensable. Additionally, as more papers that draw on alternative epistemologies are published, it will be paramount to have discussions across epistemological differences to gain a more holistic understanding of how to effectively build and bridge theory with visualization practices.

CHAPTER 8

Conclusion

Understanding how visualizations work in the world is a complicated and difficult question to answer. As researchers in this field, we work towards this aim so that we and others can use visualizations to understand and interact with data, hopefully learning something about ourselves and the world along the way. In this dissertation, we draw attention to new perspectives in visualization research that are shifting *the what* and *the how* of visualization research. Stemming from different epistemologies, our collective understanding of visualization is strengthened through these alternative methods of conducting visualization research. This dissertation contributes to the discussion by articulating a novel theory for visualization that transposes feminist entanglement epistemology to a visualization context and three case studies that highlight the productive capacity of feminist theory to change design guidelines, visualizations, and collaboration.

Entanglement theory is one of many voices in the conversation that brings alternative perspectives to visualization research. As a feminist theory, its impact goes beyond epistemology; it extends to reexamining how systems of power and privilege shape knowledge claims. The skeptical reader may ask, though: *What does this teach us about visualization?* In this dissertation, we take the stance that visualization has everything to do with epistemology and epistemology has everything to do with visualization.

As visualization researchers, our field centers on making data visible. Our decisions regarding design, data operations, and interaction give interpretable form to data. Or rather, visualizations instantiate reality as recorded by data. Visualizations become the apparatuses for knowing things about the world and perpetuate specific world views over others. This brings forward epistemological questions: *How do we know what we see is true and objective? What signals truth and objectivity?* Quickly, then, vision reinforces epistemology. If left unquestioned, this view can be confused as a window onto reality: neutral and devoid of entanglements.

Entanglements also question the objects we visualize, data. It troubles the idea of a singular truth: data about the world is fixed while the world itself remains moving. As Dumit astutely states: *"seeing and the apparatus of seeing are both interrogated and held accountable"* [48]. Accountability then draws us back to power and privilege. It is a privilege to work with data and a power to visualize it: *What do we care to show and in its shadow, what topics do we leave in the dark?* Engaging with alternative epistemologies shifts how we perceive concepts in visualization

research: it demonstrates which topics are taken for granted and which are ignored. Without this shift, we may miss the opportunity to understand how visualizations work in diverse, rich, and multi-faceted ways.

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